

PACKING $k^2 - c$ UNIT SQUARES: AN UPPER BOUND OF ORDER $k^{3/8}$ FOR THE DEFICIENCY

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ABSTRACT. For $n \geq 1$ let $s(n)$ be the side of the smallest square that contains n unit squares with pairwise disjoint interiors, and for an integer $k \geq 2$ let $c^*(k)$ be the largest integer c with $s(k^2 - c) = k$. We prove

$$c^*(k) < 43.06 k^{3/8} + 2 \cdot 10^{-5} \quad \text{for every integer } k \geq 4 \cdot 10^{26},$$

and, with much better constants for moderate k , $c^*(k) < 20.668 k^{2/5} + 0.623$ for every integer $k \geq 1.6 \cdot 10^7$; the second bound is the smaller one for all k below about $5.6 \cdot 10^{12}$. The frame is an L-shaped arrangement in the spirit of Erdős and Graham and of Chung and Graham, in which two bands of width slightly less than an integer b are filled with rows of b unit squares tilted by slightly more than $2 \arctan(1/b)$; this frame alone gives $c^*(k) \leq 8 \lceil \sqrt{k-4} \rceil - 1$ for $k \geq 6$, which we also prove. The rows are lifted, and the four ends of the bands are filled by a “staircase”: columns of squares with one common small tilt, whose lowest squares are replaced by horizontal layers so that the columns follow the rising end of the band. This gives an uncovered area $O(b^{2/3})$ per end and the exponent $2/5$. Filling the two walls of each end by chains of staircases whose tilt is lowered from block to block gives $O(b^{3/5})$ per end and the exponent $3/8$. These ideas are due to Bui; our proofs are self-contained. The constants of the $3/8$ bound are certified by interval arithmetic; its asymptotic regime cannot be tested by explicit computation, and the explicit checks test the construction at smaller sizes. Explicit packings, checked in exact rational arithmetic, also by checkers written independently of the generating program, give for example $c^*(10^5) \leq 1583$ and $c^*(10^8) \leq 24790$. Together with the lower bound of order $k^{1/3}$ of [T] (a preprint, not refereed), $k^{1/3} \ll c^*(k) \ll k^{3/8}$; the true exponent is open. The paper has not been refereed.

1. INTRODUCTION AND RESULTS

A *packing* of n unit squares in a closed square K is a family of n closed unit squares, in any orientation, contained in K and with pairwise disjoint interiors. For $n \geq 1$ let $s(n)$ be the infimum of the sides of the squares in which n unit squares can be packed. Removing squares from a packing leaves a packing, so s is non-decreasing. By area $s(k^2) \geq k$, and the grid gives $s(k^2) \leq k$; hence $s(k^2 - c) \leq k$ for $0 \leq c < k^2$. For an integer $k \geq 2$ let

$$c^*(k) := \max\{c \in \mathbb{Z} : 0 \leq c \leq k^2 - 1, s(k^2 - c) = k\},$$

the *deficiency*: removing up to $c^*(k)$ squares from the $k \times k$ grid does not allow a smaller container, while removing $c^*(k) + 1$ squares does. The wasted area of Erdős and Graham [EG] is $W(x) := x^2 - M(x)$, where $M(x)$ is the largest number of unit squares that can be packed in a square of side x . In Section 2 we record the elementary relation

$$(1.1) \quad c^*(k) = \lim_{x \rightarrow k^-} W(x) - 1,$$

so $c^*(k)$ measures the wasted area just below an integer.

Lower bounds. The author showed $c^*(k) \rightarrow \infty$ in [R], with a bound of order $\log k$; lower bounds of order $k^{1/4}$ and $k^{1/3}$ followed in [Q] and [T]. These are preprints that have not been refereed; see Section 11.

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Upper bounds. From the literature on $W(x)$ one gets $c^*(k) = O(k^{3/5})$ (preprints [Bui], [McC]); the refereed literature gives $O(k^{7/11})$ [EG] and $O(k^{(3+\sqrt{2})/7} \log k)$ [CG], and Arslanov, Mustafin and Shangitbayev [AMS] proved $c^*(k) \leq k - 1$ for $k > 12$. See Section 10.

1.1. Results. The main result is the following.

Theorem 1.1 (computer-assisted constants). *For every integer $k \geq k_1 := 4 \cdot 10^{26}$,*

$$c^*(k) < \frac{32}{5} \left(\frac{5}{3}\right)^{3/8} 15.54^{5/8} k^{3/8} + 2 \cdot 10^{-5} < 43.06 k^{3/8} + 2 \cdot 10^{-5}.$$

The proof is in Section 8. Its constants are certified by interval arithmetic (Corollary 8.3). The asymptotic regime $k \geq 4 \cdot 10^{26}$, where one end of a band has about 10^{16} pieces, cannot be tested by explicit computation; the explicit checks of Remark 8.4 test the validity of the construction and of its lemmas at smaller sizes. For moderate k the staircase alone gives a smaller bound with much better constants:

Theorem 1.2. *For every integer $k \geq k_0 := 1.6 \cdot 10^7$,*

$$c^*(k) < \frac{20}{3} \left(\frac{3}{2}\right)^{2/5} 5.03^{3/5} k^{2/5} + \frac{8}{3} \cdot 5.03 \cdot 10^{-4/3} < 20.668 k^{2/5} + 0.623.$$

The proof (Section 7) gives $c^*(k) \leq (18.89 + o(1)) k^{2/5}$ as $k \rightarrow \infty$. At $k = k_0$ the bound is 15738.5; the bound of Theorem 1.4 below is 31999 there.

Which bound to use. We keep both theorems: Theorem 1.2 is the effective bound and Theorem 1.1 the asymptotic one. For $1.6 \cdot 10^7 \leq k < 4 \cdot 10^{26}$ only Theorem 1.2 is proved; the closed forms of Theorems 1.1 and 1.2 cross at about $k = 5.6 \cdot 10^{12}$ ($43.06 k^{3/8} < 20.668 k^{2/5}$ exactly when $k^{1/40} > 43.06/20.668$). Theorem 1.2 also comes with explicit packings for moderate k (Theorem 1.6 below), while Theorem 1.1 gives the exponent $3/8$. At $k = 4 \cdot 10^{26}$ the two closed forms are about $4.07 \cdot 10^{11}$ and $9.04 \cdot 10^{11}$.

The frame of the proof is an L-shaped packing. For an integer $b \geq 2$ put $D := b^2 + 1$,

$$\gamma_0 := \frac{b^2 - 1}{D}, \quad h_0 := \frac{3b^2 - 1}{D}, \quad R(L) := \max(0, \lceil (L - h_0)\gamma_0 \rceil) \quad (L \in \mathbb{Z}),$$

and for integers $2 \leq b \leq k$

$$N(k, b) := (k - b)^2 + b(R(k) + R(k - b)).$$

Theorem 1.3. *Let $k \geq 2$ and $2 \leq b \leq k$ be integers with $(k, b) \neq (2, 2)$. Then $N(k, b)$ unit squares can be packed in a square of side less than k . Consequently, for all $2 \leq b \leq k$,*

$$c^*(k) \leq k^2 - N(k, b) - 1 < 6b + \frac{4k}{b} - 3.$$

Theorem 1.4. *For every integer $k \geq 6$,*

$$c^*(k) \leq 8\lceil \sqrt{k-4} \rceil - 1 < 8\sqrt{k-4} + 7.$$

The bound of Theorem 1.4 is the best that the packings of Theorem 1.3 give, up to an additive constant:

Proposition 1.5. *For all integers $2 \leq b \leq k$ one has $k^2 - N(k, b) > 8\sqrt{k} - 14$.*

The packings of Theorem 1.3 lose area at the four ends of the two tilted bands, about b at each end (Section 5). In Section 6 we lift the rows of each band and show that the deficiency is exactly the body cost of the bands, about $4k/b$, plus the uncovered areas of four end regions (Proposition 6.3). Section 7 fills an end region by a staircase of columns with uncovered area $O(b^{2/3})$; with $b \asymp k^{3/5}$ this gives Theorem 1.2. Section 8 replaces the two walls of the staircase by chains of staircases with slowly decreasing tilt; the uncovered area becomes $O(b^{3/5})$, and with $b \asymp k^{5/8}$ this gives Theorem 1.1. With tuned parameters the staircase construction gives explicit packings for moderate k :

Theorem 1.6 (computer-assisted). *For each row of Table 1, $k^2 - c$ unit squares can be packed in a square of side less than k . Hence $c^*(k) \leq c - 1$, as listed.*

k	b	y_0	$c = k^2 - N$	bound	Thm. 1.2	$8\lceil\sqrt{k} - 4\rceil - 1$
10^5	623	257/4	1584	$c^*(k) \leq 1583$	2067	2535
10^6	2481	653/4	3973	$c^*(k) \leq 3972$	5192	7999
10^7	9875	1663/4	10040	$c^*(k) \leq 10039$	13041	25303
$3.825 \cdot 10^7$	22086	710	16989	$c^*(k) \leq 16988$	22302	49479
10^8	39314	1041	24791	$c^*(k) \leq 24790$	32757	79999

TABLE 1. Certified packings (Theorem 1.6). The band width is $b - \varepsilon$ with $\tan(\theta/2) = b/(b^2 - 1)$, and y_0 is the lift of the first row of each band. The column “Thm. 1.2” gives the integer part of $20.668k^{2/5} + 0.623$ (Theorem 1.2 itself applies only for $k \geq 1.6 \cdot 10^7$), and the last column is the bound of Theorem 1.4. These staircase certificates are the best values we have at these k . The construction of Section 8 gives larger values here (for example $c^*(10^8) \leq 44299$, Remark 8.4); it wins only for very large k .

For comparison, the explicit bound of Wang, Dong and Li [WDL] would give about $8.1 \cdot 10^4$ at $k = 10^5$ (Section 10).

To the best of our knowledge, no upper bound of order $o(k^{3/5})$ for $c^*(k)$, or for $W(x)$ just below an integer, has been stated in the sources we searched (listed in Section 10). The ingredients of Theorem 1.3 are standard, and the staircase and the chained wall stairs are due to Bui (Section 1.2); what seems not to have been written down is their use just below an integer, with bands of width $b - \varepsilon$.

The order of $c^*(k)$. Theorem 1.1 and the lower bound of [T] give

$$k^{1/3} \ll c^*(k) \ll k^{3/8}.$$

By a theorem of Roth and Vaughan, a bound $W(x) \leq Cx^\alpha$ valid for all large real x must have $\alpha \geq 1/2$, so by (1.1) such bounds cannot give $c^*(k) = o(\sqrt{k})$ (Section 10.3); Theorems 1.1 and 1.2 use that the band width is just below an integer. These are the bounds proved here and in [T]; the true exponent of $c^*(k)$ is open: it lies between $1/3$ and $3/8$ (Section 12).

1.2. Credits. The skeleton of the packing of Theorem 1.3 (an axis-parallel square, and two bands along two of its sides filled with tilted blocks that touch both sides of the band) is the construction of Erdős and Graham [EG, Figs. 1–2]; the same L-shaped skeleton is the Type 1 packing of Chung and Graham [CG]. The block (a $1 \times b$ rectangle of b unit squares tilted by about $2\arctan(1/b)$ so that it fits in a band of width slightly less than b) is used by Arslanov, Mustafin and Shangitbayev [AMS] for small b and in a lemma for even b ; they do not chain the blocks into long bands. The staircase of Section 7, in which tilted columns of one fixed tilt keep only their upper squares and their lower squares are replaced by horizontal axis-parallel layers, so that one tilt can follow a rising ceiling, is the “primitive” construction of H. D. Bui [Bui, Section 3]. The wall filler of Section 8, in which a thin trapezoid is filled by a chain of such staircases whose tilt is lowered from block to block so that the layers close in phase, follows the ideas of [Bui, Sections 3–4] (arXiv:2508.04603v2). Our proofs of the end bounds are self-contained and do not use any argument of [Bui]; the write-up and the constants are ours.

1.3. Status. Theorems 1.3 and 1.4 and Proposition 1.5 are proved in full in Sections 2–5, Theorem 1.2 in Sections 6 and 7, and Theorem 1.1 in Sections 6 and 8. Theorem 1.2 relies on one evaluation in interval arithmetic, $\Phi(10^{-4/3}) \leq 5.0163$ (Corollary 7.4), done by the author’s program and by three independent computations; the other numerical constants were certified in the same way. The constants of Theorem 1.1 (Corollary 8.3) are certified by interval arithmetic on covers of the whole range $b \geq 10^{16}$: the author’s own program used too little slack on the wall tilts, and the constant $C_E = 15.54$ is certified by three independent full-range coverings ($C_E \leq 14.964, 14.963$ and 14.9631), with different libraries (`mpmath.iv` twice and `python-flint arb` once); the asymptotic regime cannot be tested by explicit computation, and the explicit checks test the construction at smaller b (Remark 8.4). Theorem 1.6 consists of the structural part of Section 6 and a finite part (explicit end packings), checked in exact rational arithmetic by the program that produced it and by checkers written independently from a description of the construction (Section 9). The proofs were written and checked with AI assistance and have not been reviewed by a human expert. The paper has not been refereed.

Organization. Section 2 proves (1.1) and the reduction used throughout. Section 3 treats the tilted block, Section 4 the L-shaped packing, and Section 5 the count and the proofs of Theorems 1.3, 1.4 and Proposition 1.5. Section 6 lifts the rows and proves the waste identity, Section 7 constructs the staircase end and proves Theorem 1.2, Section 8 constructs the chained wall fillers and proves Theorem 1.1, and Section 9 proves Theorem 1.6. Sections 10 and 11 compare with the literature, and Section 12 collects remarks on other end constructions and open problems.

2. THE REDUCTION AND THE WASTED AREA

For $x \geq 1$ let $M(x)$ be the largest number of unit squares that can be packed in a square of side x , and $W(x) := x^2 - M(x)$. A packing in a square of side x is also a packing in a larger square containing it, so M is non-decreasing; it is integer-valued and $M(x) \leq x^2$ by area.

Lemma 2.1. *Let $k \geq 2$ be an integer and $m(k) := \max_{1 \leq x < k} M(x)$. Then $c^*(k) = k^2 - m(k) - 1$, and $W(x) \rightarrow k^2 - m(k)$ as $x \rightarrow k^-$. In particular (1.1) holds.*

Proof. The maximum exists because M is non-decreasing, integer-valued and bounded by k^2 on $[1, k]$; and $M(x) = m(k)$ on some interval $[x_1, k]$, so $W(x) = x^2 - m(k) \rightarrow k^2 - m(k)$. Let $1 \leq n \leq k^2$. If $s(n) < k$, then by the definition of the infimum n unit squares can be packed in a square of some side $x < k$, so $n \leq M(x) \leq m(k)$. Conversely, if $n \leq m(k) = M(x)$ with $x < k$, then $s(n) \leq x < k$. Since $s(n) \leq k$ for $n \leq k^2$, we get, for $0 \leq c \leq k^2 - 1$,

$$s(k^2 - c) = k \iff k^2 - c > m(k) \iff c \leq k^2 - m(k) - 1.$$

As $1 \leq m(k) < k^2$ (one square fits in a square of side $1 < k$, and $M(x) \leq x^2 < k^2$), the largest such c is $k^2 - m(k) - 1$. $\square$

Lemma 2.2 (reduction). *If $N \geq 1$ unit squares can be packed in a square of side $S < k$, then $c^*(k) \leq k^2 - N - 1$.*

Proof. By assumption $N \leq M(S) \leq m(k)$, and Lemma 2.1 gives $c^*(k) = k^2 - m(k) - 1 \leq k^2 - N - 1$. $\square$

This is the source of the “ -1 ” in Theorems 1.3 and 1.6: a packing of N squares in a container of side less than k shows $c^*(k) \leq c - 1$ with $c := k^2 - N$. In the notation of [Q] and [T], $m(k)$ is also the largest number of unit squares in $[0, k]^2$ that are pairwise disjoint as closed sets, and $c^*(k) = W_{\min}(k) - 1$ with $W_{\min}(k) := k^2 - m(k)$ ([Q, Section 1]).

3. THE TILTED BLOCK

Fix an integer $b \geq 2$ and an angle $\theta \in (0, \pi/2)$, and write $\gamma := \cos \theta$, $\sigma := \sin \theta$, $u := (\gamma, \sigma)$ and $v := (-\sigma, \gamma)$. For $P \in \mathbb{R}^2$ and $0 \leq i \leq b-1$ let

$$Q_i(P) := \{P + \alpha u + \beta v : \alpha \in [i, i+1], \beta \in [0, 1]\}, \quad B(P) := \bigcup_{i=0}^{b-1} Q_i(P).$$

Each $Q_i(P)$ is a unit square, and $B(P)$ is a $b \times 1$ rectangle tilted by θ , with lowest vertex P . Put

$$f(\theta) := b\gamma + \sigma \quad (\text{width}), \quad h(\theta) := b\sigma + \gamma \quad (\text{height}), \quad d(\theta) := \sec \theta \quad (\text{row spacing}).$$

Lemma 3.1 (projections). *The projection of $B(P)$ to the x -axis is $[P_x - \sigma, P_x + b\gamma]$, of length $f(\theta)$, and its projection to the y -axis is $[P_y, P_y + b\sigma + \gamma]$, of length $h(\theta)$.*

Proof. The coordinates $x = P_x + \alpha\gamma - \beta\sigma$ and $y = P_y + \alpha\sigma + \beta\gamma$ are linear on $[0, b] \times [0, 1] \ni (\alpha, \beta)$, and $\gamma, \sigma > 0$. So x is smallest at $(0, 1)$ and largest at $(b, 0)$, and y is smallest at $(0, 0)$ and largest at $(b, 1)$. $\square$

Lemma 3.2 (width). *Let $\theta_0 := 2 \arctan(1/b)$. Then $0 < \theta_0 < \pi/2$,*

$$\cos \theta_0 = \gamma_0 = \frac{b^2 - 1}{D}, \quad \sin \theta_0 = \frac{2b}{D}, \quad f(\theta_0) = b, \quad h(\theta_0) = h_0 = \frac{3b^2 - 1}{D}, \quad d(\theta_0) = d_0 := \frac{D}{b^2 - 1},$$

with $D = b^2 + 1$, and $f(\theta) < b$ for every $\theta \in (\theta_0, \pi/2]$.

Proof. With $t = \tan(\theta_0/2) = 1/b$ the half-angle formulas give $\cos \theta_0 = (1 - t^2)/(1 + t^2) = (b^2 - 1)/D$ and $\sin \theta_0 = 2t/(1 + t^2) = 2b/D$; hence $f(\theta_0) = (b(b^2 - 1) + 2b)/D = b$ and $h(\theta_0) = (2b^2 + b^2 - 1)/D$. As $b \geq 2$, $\arctan(1/b) < \pi/4$, so $\theta_0 < \pi/2$. With $\varphi := \arctan(1/b)$ we have $f(\theta) = \sqrt{b^2 + 1} \cos(\theta - \varphi)$. For $\theta \in [\theta_0, \pi/2]$, $\theta - \varphi \in [\varphi, \pi/2 - \varphi] \subset [0, \pi]$, where $\cos$ is strictly decreasing; so $f(\theta) < f(\theta_0) = b$ for $\theta > \theta_0$. $\square$

Lemma 3.3 (stacking). *Let $P_0 \in \mathbb{R}^2$ and $P_j := P_0 + (0, j \sec \theta)$ for $j \in \mathbb{Z}$. If $(i, j) \neq (i', j')$, then $Q_i(P_j)$ and $Q_{i'}(P_{j'})$ have disjoint interiors. More precisely, in the coordinates $\alpha' = u \cdot (z - P_0)$, $\beta' = v \cdot (z - P_0)$,*

$$Q_i(P_j) = [i + j \tan \theta, i + 1 + j \tan \theta] \times [j, j + 1].$$

Proof. The map $z \mapsto (u \cdot (z - P_0), v \cdot (z - P_0))$ is a rigid motion. Since $v \cdot (0, j \sec \theta) = j$ and $u \cdot (0, j \sec \theta) = j \tan \theta$, the square $Q_i(P_j)$ has the stated form. If $j \neq j'$ the β' -intervals have disjoint interiors; if $j = j'$ and $i \neq i'$ the α' -intervals do. $\square$

Thus consecutive rows touch along their long sides: the v -component of the shift $(0, \sec \theta)$ is exactly 1.

Lemma 3.4 (one band). *Let $a \in \mathbb{R}$, $L \geq 0$, $\theta \in (0, \pi/2)$ and an integer $R \geq 0$ with $R = 0$ or $(R-1)d(\theta) + h(\theta) \leq L$. Then the bR squares $Q_i(P_j)$, $0 \leq i < b$, $0 \leq j < R$, with $P_j = (a + \sigma, j \sec \theta)$, have pairwise disjoint interiors and lie in $[a, a + f(\theta)] \times [0, L]$.*

Proof. By Lemma 3.1 the row $B(P_j)$ has x -projection $[a, a + b\gamma + \sigma]$ and y -projection $[jd, jd + h] \subset [0, (R-1)d + h] \subset [0, L]$. Disjointness is Lemma 3.3. $\square$

4. THE L-SHAPED PACKING

Let $k \geq 2$ and $2 \leq b \leq k$ be integers and $\theta \in (\theta_0, \pi/2)$. By Lemma 3.2,

$$\varepsilon := b - f(\theta) > 0, \quad S := k - \varepsilon = k - b + f(\theta) < k.$$

Let $R_1, R_2 \geq 0$ be integers satisfying the *length conditions*

$$(L1) \quad R_1 = 0 \quad \text{or} \quad (R_1 - 1)d(\theta) + h(\theta) \leq S, \quad (L2) \quad R_2 = 0 \quad \text{or} \quad (R_2 - 1)d(\theta) + h(\theta) \leq k - b.$$

The packing consists of three parts (Figure 1 of [EG] and the Type 1 packing of [CG] have the same skeleton).

- (1) The *grid* $\mathcal{G}$: the $(k-b)^2$ squares $[i, i+1] \times [j, j+1]$, $0 \leq i, j \leq k-b-1$, in $G := [0, k-b]^2$.
- (2) The *right band* $\mathcal{A}$: the bR_1 squares of Lemma 3.4 with $a = k-b$, $L = S$, $R = R_1$; they lie in $A := [k-b, S] \times [0, S]$, since $a + f(\theta) = S$.
- (3) The *top band* $\mathcal{T}$: the image under the reflection $\iota(x, y) := (y, x)$ of the bR_2 squares of Lemma 3.4 with $a = k-b$, $L = k-b$, $R = R_2$. Before reflection they lie in $[k-b, S] \times [0, k-b]$, so $\mathcal{T}$ lies in $T := [0, k-b] \times [k-b, S]$.

Lemma 4.1. *The $N = (k-b)^2 + b(R_1 + R_2)$ squares of $\mathcal{G} \cup \mathcal{A} \cup \mathcal{T}$ form a packing in $[0, S]^2$, and $S < k$.*

Proof. Since $f(\theta) > 0$ we have $k-b < S$, so $G, A, T \subset [0, S]^2$; and $S < k$. The interiors $(0, k-b)^2$, $(k-b, S) \times (0, S)$ and $(0, k-b) \times (k-b, S)$ of G, A, T are pairwise disjoint (G and A are separated by $x = k-b$, G and T by $y = k-b$, and A and T by $x = k-b$); the corner $[k-b, S]^2$ belongs to A only. The interior of a square contained in one of these regions lies in the interior of that region, so squares from different parts have disjoint interiors. Inside $\mathcal{G}$ this is clear, inside $\mathcal{A}$ it is Lemma 3.4, and inside $\mathcal{T}$ it is Lemma 3.4 together with the fact that ι is an isometry. $\square$

If $k = b$, the grid is empty, T is degenerate and $R_2 = 0$; the lemma remains true.

4.1. An explicit angle. For an integer L let $R(L)$ be as in Section 1. Then $R(L)$ is the number of integers $j \geq 0$ with $jd_0 + h_0 < L$: if $L \leq h_0$ both are 0; otherwise the condition is $j < (L - h_0)/d_0 = (L - h_0)\gamma_0$, and the number of such $j \geq 0$ is $\lceil (L - h_0)\gamma_0 \rceil$.

Lemma 4.2. *Let $M := 14kb^2 + 1$, $p := M$, $q := bM - 1$ and $\theta := 2 \arctan(p/q)$. Then $\theta \in (\theta_0, \pi/2)$, and $R_1 := R(k)$, $R_2 := R(k-b)$ satisfy (L1) and (L2). Moreover $\cos \theta = (q^2 - p^2)/(q^2 + p^2)$ and $\sin \theta = 2pq/(p^2 + q^2)$ are rational, so all vertices of the packing have rational coordinates.*

Proof. (i) We have $p/q > 1/b$ (as $bM > bM - 1$) and $p/q < 1$ (as $(b-1)M > 1$), so $\theta_0 < \theta < \pi/2$. Since the derivative of $\arctan$ is at most 1,

$$\theta - \theta_0 = 2 \left(\arctan \frac{p}{q} - \arctan \frac{1}{b} \right) \leq 2 \left(\frac{M}{bM-1} - \frac{1}{b} \right) = \frac{2}{b(bM-1)} \leq \frac{1}{7kb^4} =: \tau,$$

because $bM - 1 = 14kb^3 + b - 1 \geq 14kb^3$.

(ii) *Slack.* Let L be an integer with $R := R(L) \geq 1$, and $g(L) := L - h_0 - (R-1)d_0$. Then $g(L) > 0$ by the characterization of $R(L)$ above, and

$$(b^2 - 1)Dg(L) = (b^4 - 1)L - (b^2 - 1)(3b^2 - 1) - (R-1)D^2$$

is an integer; hence $g(L) \geq 1/(b^4 - 1)$.

(iii) *Derivatives.* For $b \geq 2$, $\gamma_0 \geq 3/5$, and $\tau \leq 1/224$; so $\cos \geq 3/5 - \tau \geq 1/2$ on $[\theta_0, \theta_0 + \tau]$. There $d' = \sin / \cos^2 \leq 4$, $h' = b \cos - \sin \leq b$ and $|f'| = |\cos - b \sin| \leq \sqrt{b^2 + 1} \leq b + 1$.

(iv) If $R_1 = R(k) \geq 1$ (otherwise there is nothing to prove), then by the mean value theorem, using $f(\theta_0) = b$,

$$\begin{aligned} (R_1 - 1)d(\theta) + h(\theta) - S &= [(R_1 - 1)d_0 + h_0 - k] + (R_1 - 1)(d(\theta) - d_0) \\ &\quad + (h(\theta) - h_0) + (f(\theta_0) - f(\theta)) \\ &\leq -g(k) + (\theta - \theta_0)(4(R_1 - 1) + 2b + 1). \end{aligned}$$

Here $R_1 - 1 < (k - h_0)\gamma_0 < k$ and $b \leq k$, so the last factor is at most $7k$, and the right side is at most $-1/(b^4 - 1) + 7k\tau = -1/(b^4 - 1) + 1/b^4 < 0$. This is (L1). Similarly, if $R_2 \geq 1$, $(R_2 - 1)d(\theta) + h(\theta) - (k - b) \leq -g(k - b) + \tau(4k + b) \leq -1/(b^4 - 1) + 5/(7b^4) < 0$, which is (L2). $\square$

Lemma 4.3 (maximality within the family). *If $\theta \in (\theta_0, \arctan b)$, then every pair R_1, R_2 satisfying (L1) and (L2) has $R_1 \leq R(k)$ and $R_2 \leq R(k - b)$. For $\theta \in (\theta_0, \theta_0 + \tau]$ equality is possible.*

Proof. Note $\tan \theta_0 = 2b/(b^2 - 1) < b$, so the interval is non-empty. On it $d' > 0$ and $h' = b \cos \theta - \sin \theta > 0$, so d and h are strictly increasing. If $R_1 \geq R(k) + 1$, then $(R_1 - 1)d_0 + h_0 \geq k$ by the characterization of $R(k)$, hence $(R_1 - 1)d(\theta) + h(\theta) > k > S$, contradicting (L1). The top band is the same with $k - b$ in place of S and k . The proof of Lemma 4.2 works for every $\theta \in (\theta_0, \theta_0 + \tau]$. $\square$

So $N(k, b)$ is the exact number of squares of the packings of this section with θ close to θ_0 and with the rows of each band starting at the bottom of the band. Remark 5.3 shows that the family can be enlarged.

5. THE COUNT: PROOFS OF THE BOUNDS OF ORDER $\sqrt{k}$

Proof of Theorem 1.3. By Lemmas 4.2 and 4.1, $N(k, b)$ unit squares can be packed in a square of side $S < k$. If $k > b$ then $N(k, b) \geq (k - b)^2 \geq 1$; if $k = b$ then $N(k, b) = bR(k)$, and $R(k) = 0$ only if $k \leq h_0 < 3$, that is $k = b = 2$. So $N(k, b) \geq 1$ for $(k, b) \neq (2, 2)$, and Lemma 2.2 gives $c^*(k) \leq k^2 - N(k, b) - 1$; for $(k, b) = (2, 2)$ this bound is $c^*(2) \leq 3$, which holds by definition.

For the second inequality we compute the count exactly. Put

$$\eta := \frac{2k - 10}{D} + \frac{8}{D^2}, \quad \eta' := \eta - \frac{2b}{D}.$$

We claim

$$(5.1) \quad k^2 - N(k, b) \leq 6b + b(\lfloor \eta \rfloor + \lfloor \eta' \rfloor), \quad \text{with equality if } k - b \geq 2.$$

For an integer L , using $h_0 = 3 - 4/D$ and $\gamma_0 = 1 - 2/D$,

$$(L - h_0)\gamma_0 = L - 3 + \frac{4}{D} - \frac{2L}{D} + \frac{6}{D} - \frac{8}{D^2} = L - 3 - \eta_L, \quad \eta_L := \frac{2L - 10}{D} + \frac{8}{D^2}.$$

Since L is an integer, $\lceil L - 3 - \eta_L \rceil = L - 3 - \lfloor \eta_L \rfloor$, so $R(L) = \max(0, L - 3 - \lfloor \eta_L \rfloor) \geq L - 3 - \lfloor \eta_L \rfloor$. Here $\eta_k = \eta$ and $\eta_{k-b} = \eta'$. Hence

$$k^2 - N(k, b) = 2kb - b^2 - b(R(k) + R(k - b)) \leq 2kb - b^2 - b(2k - b - 6 - \lfloor \eta \rfloor - \lfloor \eta' \rfloor) = 6b + b(\lfloor \eta \rfloor + \lfloor \eta' \rfloor).$$

If $L \geq 2$, then $L - h_0 > -1$ and $0 < \gamma_0 < 1$, so $(L - h_0)\gamma_0 > -1$ and the ceiling is at least 0; the maximum is then not active. This applies to $L = k$ always and to $L = k - b$ when $k - b \geq 2$, which gives the equality case.

Now $\lfloor \eta \rfloor + \lfloor \eta' \rfloor \leq 2\eta - 2b/D$, and therefore

$$k^2 - N(k, b) \leq 6b + \frac{(4k - 20)b}{D} + \frac{16b}{D^2} - \frac{2b^2}{D} = 6b + \frac{4kb}{D} - 2 + \frac{2 - 20b + 16b/D}{D}.$$

Here $4kb/D < 4k/b$, and for $b \geq 2$ we have $16b/D \leq 32/5 < 7$, so $2 - 20b + 16b/D < 9 - 20b < 0$. Hence $k^2 - N(k, b) < 6b + 4k/b - 2$, and $c^*(k) \leq k^2 - N(k, b) - 1 < 6b + 4k/b - 3$. $\square$

Corollary 5.1. *For every integer $k \geq 6$, $c^*(k) < 4\sqrt{6}\sqrt{k} \approx 9.798\sqrt{k}$.*

Proof. The function $g(x) = 6x + 4k/x$ has its minimum $g(x^*) = 4\sqrt{6}\sqrt{k}$ at $x^* = \sqrt{2k/3}$. Let $b = \lceil x^* \rceil = x^* + \delta$ with $0 \leq \delta < 1$. Using $4k/x^{*2} = 6$,

$$g(b) - g(x^*) = 6\delta - \frac{4k\delta}{x^*(x^* + \delta)} = \frac{6\delta^2}{x^* + \delta} < \frac{6}{x^*} \leq 3,$$

because $x^* \geq 2$ for $k \geq 6$. Also $2 \leq b \leq x^* + 1 \leq k$. Theorem 1.3 gives $c^*(k) < g(b) - 3 < 4\sqrt{6}\sqrt{k}$. $\square$

Proof of Theorem 1.4. Let $b := \lceil \sqrt{k - 4} \rceil$. Then $b \geq \lceil \sqrt{2} \rceil = 2$ and $b \leq \sqrt{k - 4} + 1 \leq k$. Moreover $D = b^2 + 1 \geq k - 3 > 0$ and $2k - 10 > 0$, so

$$\eta \leq \frac{2k - 10}{k - 3} + \frac{8}{(k - 3)^2} = 2 - \frac{4(k - 3) - 8}{(k - 3)^2} < 2,$$

since $k - 3 > 2$. Hence $\lfloor \eta \rfloor \leq 1$ and $\lfloor \eta' \rfloor \leq \lfloor \eta \rfloor \leq 1$, and (5.1) gives $k^2 - N(k, b) \leq 8b$. Since $(k, b) \neq (2, 2)$, Theorem 1.3 gives $c^*(k) \leq 8b - 1 < 8\sqrt{k - 4} + 7$. $\square$

For $k > 15$ Theorem 1.4 is stronger than Corollary 5.1, since $8\sqrt{k} + 7 < 4\sqrt{6}\sqrt{k}$ for $\sqrt{k} > 7/(4\sqrt{6} - 8) \approx 3.89$.

Proof of Proposition 1.5. If $k - b \geq 2$, equality holds in (5.1), and $\lfloor z \rfloor > z - 1$ gives

$$k^2 - N(k, b) > 6b + b\left(2\eta - \frac{2b}{D} - 2\right) = 4b + \frac{(4k - 20)b}{D} + \frac{16b}{D^2} - \frac{2b^2}{D} \geq 4b + \frac{4kb}{D} - 12,$$

using $16b/D^2 \geq 0$, $2b^2/D \leq 2$ and $20b/D \leq 10$. With $x := b + 1/b$ we have $kb/D = k/x$ and $4b = 4x - 4/b \geq 4x - 2$, so $k^2 - N(k, b) > 4x + 4k/x - 14 \geq 8\sqrt{k} - 14$ by the inequality of arithmetic and geometric means. If $b \in \{k - 1, k\}$, then $R(k - b) \in \{R(0), R(1)\} = \{0\}$ (as $h_0 \geq 11/5$) and $R(k) \leq \lceil k - h_0 \rceil \leq k - 2$, so $N(k, b) \leq 1 + k(k - 2)$ and $k^2 - N(k, b) \geq 2k - 1$; and $2k - 1 - (8\sqrt{k} - 14) = 2(\sqrt{k} - 2)^2 + 5 > 0$. $\square$

Hence the best bound obtainable from Theorem 1.3 satisfies $8\sqrt{k} - 15 < \min_b(k^2 - N(k, b) - 1) \leq 8\lceil\sqrt{k} - 4\rceil - 1$.

Remark 5.2 (where $8\sqrt{k}$ comes from). The exact count (5.1) is $6b + b(\lfloor \eta \rfloor + \lfloor \eta' \rfloor)$ with $\eta \approx 2k/b^2$. The estimate $\lfloor z \rfloor \leq z$ gives $6b + 4k/b$, minimized at about $9.80\sqrt{k}$; but for $b = \lceil\sqrt{k} - 4\rceil$ and $k \geq 23$ both floors equal 1 while η is just below 2, and the count is $8b$. (For $k \geq 31$: $\eta < 2$ by the proof of Theorem 1.4, and with $s := \sqrt{k} - 4$, $b < s + 1$ gives $b^2 + 2b < k - 1 + 4s$, so $D\eta' > 2k - 10 - 2b > b^2 + 1 = D$ as soon as $k - 10 \geq 4s$, which holds for $k \geq 31$; hence $1 < \eta' \leq \eta < 2$. For $23 \leq k \leq 30$ this was checked exactly, as was the whole range $23 \leq k \leq 2 \cdot 10^5$; $k = 22$ fails. Nothing else in the paper uses this.) Geometrically (Section 6), each band loses about $2L/b$ along its length L , because the row spacing $\sec \theta$ exceeds 1 by about $2/b^2$, and about b at each of its two ends, where the first and the last row cross the band with a rise of about 2. The total $4k/b + 4b$ is at least $8\sqrt{k}$.

Remark 5.3 (a larger family). Lemma 4.3 concerns the family in which the top band stays in $x \leq k - b$. One more row in the top band protrudes into the corner $[k - b, S]^2$, but often fits into the empty wedge above the last row of the right band. An exact computer check (all $6 \leq k \leq 30$ with $k - b \geq 2$) found 285 of 375 such enlarged packings valid; for example $k = 12$, $b = 3$ gives 126 squares instead of $N(12, 3) = 123$. We have no general proof, and this remark is not used.

6. LIFTED ROWS AND THE WASTE IDENTITY

6.1. Lifted rows and end regions. Throughout this section $b \geq 2$, $\theta \in (\theta_0, \pi/2)$, $\gamma = \cos \theta$, $\sigma = \sin \theta$, and $w := f(\theta) = b\gamma + \sigma < b$, $h := h(\theta)$, $d := \sec \theta$. A band is described in its own *strip frame*: the strip is $\Sigma := [0, w] \times [0, L_b]$, where $L_b > 0$ is the length of the band. Given a *lift* $y_0 \geq 0$ and an integer $R \geq 1$, the rows are $B(P_j)$, $0 \leq j < R$, with

$$P_j := (\sigma, y_0 + jd), \quad \text{and we put} \quad y_1 := L_b - h - (R - 1)d - y_0.$$

We assume $y_1 \geq 0$. For $y \geq 0$ the *end region* is

$$\text{Reg}(y) := \{(x', y') : 0 \leq x' \leq w, \ 0 \leq y' \leq y + (x' - \sigma) \tan \theta\},$$

and $\rho(x', y') := (w - x', L_b - y')$ is the rotation by π about the centre of Σ . The upper side of $\text{Reg}(y_0)$ is the line through the lower side of the first row $B(P_0)$; on $[0, \sigma]$ the first row is higher than this line, so $\text{Reg}(y_0)$ is slightly smaller than the part of Σ below the first row.

Lemma 6.1. *Let $\mathcal{C}_0$ and $\mathcal{C}_1$ be finite families of unit squares with pairwise disjoint interiors, contained in $\text{Reg}(y_0)$ and $\text{Reg}(y_1)$ respectively. Then the bR squares $Q_i(P_j)$ ($0 \leq i < b$, $0 \leq j < R$), the squares of $\mathcal{C}_0$ and the squares of $\rho(\mathcal{C}_1)$ form a packing in Σ .*

Proof. Containment. By Lemma 3.1 the row $B(P_j)$ has x -projection $[0, w]$ and y -projection $[y_0 + jd, y_0 + jd + h] \subset [y_0, L_b - y_1] \subset [0, L_b]$. On $\text{Reg}(y)$ we have $y' \leq y + (w - \sigma) \tan \theta = y + b\sigma \leq y + h$; since $L_b = y_0 + y_1 + (R - 1)d + h$ is at least $y_0 + h$ and at least $y_1 + h$, both end regions lie in Σ , and $\rho(\Sigma) = \Sigma$.

Separation. Let $\beta'(z) := v \cdot (z - P_0)$. By Lemma 3.3 (with this P_0), every row square lies in $\{0 \leq \beta' \leq R\}$. The upper side of $\text{Reg}(y_0)$ is the line through P_0 with direction u , and $\beta'(x', y') = -\sigma(x' - \sigma) + \gamma(y' - y_0) \leq 0$ is equivalent to $y' \leq y_0 + (x' - \sigma) \tan \theta$; so $\text{Reg}(y_0) \subset \{\beta' \leq 0\}$. Next, the highest vertex of the last row is $P_{R-1} + bu + v = (b\gamma, y_0 + (R - 1)d + h)$, and

$$(6.1) \quad \rho(P_{R-1} + bu + v) = (w - b\gamma, L_b - y_0 - (R - 1)d - h) = (\sigma, y_1) =: P'_0.$$

As above, $\text{Reg}(y_1) \subset \{z : v \cdot (z - P'_0) \leq 0\}$. The linear part of ρ is $-\text{id}$, so for $z \in \text{Reg}(y_1)$

$\beta'(\rho(z)) = v \cdot (\rho(P'_0) - P_0) - v \cdot (z - P'_0) = v \cdot (P_{R-1} + bu + v - P_0) - v \cdot (z - P'_0) \geq (R - 1) + 1 = R$, using $v \cdot (P_{R-1} - P_0) = R - 1$, $v \cdot u = 0$ and $v \cdot v = 1$. Hence $\rho(\text{Reg}(y_1)) \subset \{\beta' \geq R\}$. The three sets $\{\beta' \leq 0\}$, $\{0 \leq \beta' \leq R\}$ and $\{\beta' \geq R\}$ have pairwise disjoint interiors; inside each group the interiors are disjoint by Lemma 3.3 and by assumption (ρ is an isometry). $\square$

Corollary 6.2. *Let $k > b$ be an integer, $S := k - b + w$, and $y_0 \geq 0$. For the right band put $L_b := S$ and for the top band $L_b := k - b$, and suppose that for each band an integer $R \geq 1$ is given with $y_1 \geq 0$. Let $\mathcal{C}_0, \mathcal{C}_1$ be as in Lemma 6.1 for each band. Map the right band into the plane by $(x', y') \mapsto (k - b + x', y')$, and the top band by $(x', y') \mapsto \iota(k - b + x', y') = (y', k - b + x')$. Together with the grid $\mathcal{G}$ these squares form a packing in $[0, S]^2$, with $S < k$ and*

$$N = (k - b)^2 + b(R_{\text{right}} + R_{\text{top}}) + \sum_{4 \text{ ends}} |\mathcal{C}|.$$

Proof. The strips are mapped onto A and T of Section 4. Lemma 6.1 gives a packing inside each of A and T , and the separation of G, A, T is as in the proof of Lemma 4.1. $\square$

6.2. The waste identity. For $y \geq 0$ let

$$A_0(y) := yw + \frac{1}{2}\gamma\sigma(1 + b^2).$$

If $y = y_0$, this is the area of the part of Σ below the lower boundary of the first row: that boundary is the segment from $(0, y_0 + \gamma)$ to (σ, y_0) followed by the segment from (σ, y_0) to $(w, y_0 + b\sigma)$, and

$$\int_0^\sigma \left(y_0 + \gamma - \frac{\gamma}{\sigma}x\right)dx + \int_\sigma^w (y_0 + (x - \sigma) \tan \theta)dx = y_0\sigma + \frac{\gamma\sigma}{2} + y_0b\gamma + \frac{b^2\gamma\sigma}{2} = A_0(y_0).$$

For a finite family $\mathcal{C}$ in $\text{Reg}(y)$ we write $E(y) := A_0(y) - |\mathcal{C}|$, the *end cost*. It is the uncovered area of the part of Σ below the first row (for $y = y_0$), resp. above the last row (for $y = y_1$, after applying ρ).

Proposition 6.3 (waste identity). *In the situation of Lemma 6.1, with $E(y_0) = A_0(y_0) - |\mathcal{C}_0|$ and $E(y_1) = A_0(y_1) - |\mathcal{C}_1|$,*

$$wL_b - bR - |\mathcal{C}_0| - |\mathcal{C}_1| = (R - 1) \tan \theta + E(y_0) + E(y_1).$$

In the situation of Corollary 6.2,

$$k^2 - N = (k^2 - S^2) + \sum_{2 \text{ bands}} [(R - 1) \tan \theta + E(y_0) + E(y_1)].$$

Proof. Split Σ into the part below the lower boundary of $B(P_0)$ (area $A_0(y_0)$), the part between the lower boundary of $B(P_0)$ and the upper boundary of $B(P_{R-1})$, and the part above the upper boundary of $B(P_{R-1})$. The lower boundary of $B(P_j)$ is that of $B(P_0)$ shifted up by jd , and each row has x -projection $[0, w]$; so the middle part has area $(R - 1)dw + b$, where b is the area of $B(P_{R-1})$. By (6.1), ρ maps $B(P_{R-1})$ onto the rectangle $B(P'_0)$ (a rotation by π maps a rectangle with sides

along u and v onto such a rectangle, and the highest vertex onto the lowest one), so the top part has area $A_0(y_1)$. Hence $wL_b = A_0(y_0) + A_0(y_1) + (R-1)dw + b$, and since $dw = b + \tan \theta$,

$$wL_b - bR - |\mathcal{C}_0| - |\mathcal{C}_1| = E(y_0) + E(y_1) + (R-1)(dw - b) = E(y_0) + E(y_1) + (R-1)\tan \theta.$$

For the second identity, $[0, S]^2$ is the union of G , A and T with disjoint interiors, G is covered by the grid, and A , T are the images of the two strips. $\square$

The term $(R-1)\tan \theta$ is the *body cost* of a band. Since $\tan \theta \approx 2/b$ and $R \approx L_b$, the two bands cost about $4k/b$; the four ends cost $4E$, and $k^2 - S^2 = (b-w)(k+S)$ is tiny when θ is close to θ_0 . For the packing of Theorem 1.3 ($y_0 = 0$, no end squares) $E(0) = \frac{1}{2}\gamma\sigma(1+b^2) \approx b$; this is the end cost $\approx b$ of Remark 5.2. Any packing of $\text{Reg}(y)$ with end cost $E = o(b)$, for lifts y in an interval of length $\sec \theta$, makes the total cost $4k/b + o(b)$; Section 7 constructs such packings with $E = O(b^{2/3})$.

7. THE STAIRCASE END AND THE EXPONENT 2/5

In this section we prove Theorem 1.2. Section 7.1 packs a general trapezoid of the form $\text{Reg}(y)$ by a staircase of columns with one common tilt (Lemma 7.3); Section 7.2 specializes to the band ends, and Section 7.3 assembles the packing. The idea of replacing the lower squares of tilted columns by horizontal layers is that of [Bui, Section 3] (Section 1.2).

7.1. Columns, layers and the end lemma. Fix reals $w > 0$, $\mu > 0$, $\sigma_0 \geq 0$ and y with $g_0 := y - \sigma_0\mu > 1$, and put

$$g(X) := y + (X - \sigma_0)\mu, \quad \text{Reg} := \{(X, Y) : 0 \leq X \leq w, 0 \leq Y \leq g(X)\},$$

$\bar{g} := g(w)$ and $\Delta := \bar{g} - g_0 = w\mu$. Reg is a convex trapezoid of area $\int_0^w g$, and g is increasing with $g(0) = g_0$. For a packing in Reg let U be the area of Reg minus the number of unit squares of the packing (the *uncovered area*).

Tilt notation. For $t \in (0, 1)$ let $\alpha := 2 \arctan t$, so that

$$\cos \alpha = \frac{1-t^2}{1+t^2}, \quad \sin \alpha = \frac{2t}{1+t^2}, \quad \tan \alpha = \frac{2t}{1-t^2}, \quad \sec \alpha = \frac{1+t^2}{1-t^2}$$

are rational if t is. Put $u_\alpha := (\cos \alpha, \sin \alpha)$, $v_\alpha := (-\sin \alpha, \cos \alpha)$ and $\lambda(X, Y) := X + Y \tan \alpha$ (the *slab coordinate*). We use $\sec \alpha \cos \alpha = 1$, $\sec \alpha \sin \alpha = \tan \alpha$, $\sin \alpha \leq \tan \alpha$, $\sin \alpha \leq 2t$, $1 - \cos \alpha = 2t^2/(1+t^2) \leq 2t^2$, and that $\tan \alpha$ and $\sec \alpha$ are increasing in t .

Columns. For a real ξ and integers $J \geq 0$, $n \geq 1$ put $P := (\xi - J \tan \alpha, J)$ and

$$K(\xi, J, n) := \{P + a u_\alpha + e v_\alpha : a \in [0, 1], e \in [0, n]\},$$

the union of the n unit squares $\{P + a u_\alpha + e v_\alpha : a \in [0, 1], e \in [i, i+1]\}$, $0 \leq i < n$: a $1 \times n$ rectangle tilted by α and leaning to the left, with lowest vertex P .

Lemma 7.1 (columns). *Every point (X, Y) of $K(\xi, J, n)$ satisfies*

- (i) $\xi \leq \lambda(X, Y) \leq \xi + \sec \alpha$;
- (ii) $J \leq Y \leq J + n \cos \alpha + \sin \alpha$;
- (iii) $X \geq \xi - J \tan \alpha - n \sin \alpha \geq \xi - (J + n) \tan \alpha$;
- (iv) $X \leq \xi - J \tan \alpha + \cos \alpha \leq \xi + \sec \alpha$.

Proof. For $z = P + a u_\alpha + e v_\alpha$ we have $X = \xi - J \tan \alpha + a \cos \alpha - e \sin \alpha$ and $Y = J + a \sin \alpha + e \cos \alpha$. Hence $\lambda = \xi - J \tan \alpha + a \cos \alpha - e \sin \alpha + J \tan \alpha + a \sin \alpha \tan \alpha + e \cos \alpha \tan \alpha = \xi + a(\cos \alpha + \sin \alpha^2 / \cos \alpha) = \xi + a \sec \alpha$, which is (i). (ii), (iii) and (iv) follow from $a \in [0, 1]$, $e \in [0, n]$, $\sin \alpha, \cos \alpha \geq 0$, $\sin \alpha \leq \tan \alpha$ and $\cos \alpha \leq 1 \leq \sec \alpha$. $\square$

Lemma 7.2 (the tilt). *Let m be an integer and g_0 a real number with $1 < g_0 < m$. Put $D_m := m + g_0$, $\delta := m - g_0 > 0$, $F(t) := D_m t^2 - 2t - \delta$ and*

$$t^* := \frac{1 + \sqrt{1 + D_m \delta}}{D_m}.$$

Then $t^ \in (0, 1)$, and for $t \in (0, 1)$: $m \cos \alpha + \sin \alpha \leq g_0$ if and only if $F(t) \geq 0$, if and only if $t \geq t^*$. Moreover, if $g_0 \geq \nu > 0$ and $\delta \leq \bar{\delta}$, then $t^* \leq (1 + \sqrt{1 + 2\nu\bar{\delta}})/(2\nu)$.*

Proof. With $\cos \alpha, \sin \alpha$ as above, $(1 + t^2)(m \cos \alpha + \sin \alpha - g_0) = m(1 - t^2) + 2t - g_0(1 + t^2) = -F(t)$. F is a convex quadratic with $F(0) = -\delta < 0$, so $F < 0$ on $[0, t^*)$ and $F > 0$ on (t^*, ∞) , where t^* is its positive root; $F(1) = 2g_0 - 2 > 0$ gives $t^* < 1$. For the bound, $f(D', e) := (1 + \sqrt{1 + D'e})/D'$ is increasing in $e \geq 0$ and decreasing in $D' > 0$, because

$$\frac{\partial f}{\partial D'} = \frac{D'e/(2\sqrt{1 + D'e}) - 1 - \sqrt{1 + D'e}}{D'^2} \quad \text{and} \quad \frac{D'e}{2\sqrt{1 + D'e}} < \frac{\sqrt{1 + D'e}}{2}.$$

Since $D_m = m + g_0 > 2g_0 \geq 2\nu$ and $\delta \leq \bar{\delta}$, $t^* = f(D_m, \delta) \leq f(2\nu, \bar{\delta})$. $\square$

The construction. Let $Q \geq 1$ be an integer (the tilt grid).

- (S1) *Column length:* $m := \lfloor \bar{g} \rfloor + 1$. So $m > \bar{g} \geq g_0$, and $\delta := m - g_0 \in (\Delta, \Delta + 1]$.
- (S2) *Tilt:* $t := \lceil t^* Q \rceil / Q$ with $t^* = t^*(m, g_0)$ of Lemma 7.2, so $t \in [t^*, t^* + 1/Q]$; we require $t < 1$.
Let $\alpha := 2 \arctan t$; this one tilt is used for all columns.
- (S3) *Slabs:* $\xi_i := m \tan \alpha + i \sec \alpha$ for $i = 0, 1, 2, \dots$, and $I := \#\{i \geq 0 : \xi_i + \sec \alpha \leq w\}$; we require $I \geq 1$, that is $m \tan \alpha + \sec \alpha \leq w$. Then $w - \sec \alpha < \xi_I \leq w$.
- (S4) *Staircase:* for $0 \leq i < I$ let $\Gamma_i := g(\xi_i - m \tan \alpha)$, $H(J) := J + (m - J) \cos \alpha + \sin \alpha$,

$$J_i := \left\lfloor \frac{\Gamma_i - m \cos \alpha - \sin \alpha}{1 - \cos \alpha} \right\rfloor, \quad n_i := m - J_i,$$

and the column $K_i := K(\xi_i, J_i, n_i)$ (n_i squares with base at height J_i).

- (S5) *Layers:* for integers $j \geq 0$ let $p(j) := \#\{i < I : J_i \leq j\}$ and $a_j := \xi_{p(j)} - j \tan \alpha$, and let j_1 be the largest $j \in \{-1, 0, 1, \dots, \lfloor \bar{g} \rfloor\}$ such that $g(a_{j'}) \geq j' + 1$ for all integers $0 \leq j' \leq j$. For $0 \leq j \leq j_1$ put $L_j := \lfloor w - a_j \rfloor$ and, if $L_j \geq 1$, the layer $\Lambda_j := [a_j, a_j + L_j] \times [j, j + 1]$ (L_j axis-parallel unit squares in a row).
- (S6) *Left wedge:* for $0 \leq r \leq \lfloor g_0 \rfloor - 1$ put $l_r := \lfloor (m - r - 1) \tan \alpha \rfloor$ and, if $l_r \geq 1$, the row $\Omega_r := [0, l_r] \times [r, r + 1]$ (l_r axis-parallel unit squares).

All columns have the same tilt and the same total length m : column i keeps its top n_i squares tilted, and its J_i lowest squares are replaced by the layers $j < J_i$. Each layer raises the top of a column by exactly $1 - \cos \alpha$, which is how a fixed tilt follows the rising ceiling ($J_0 \leq J_1 \leq \dots$, a staircase). The left wall is handled by the rows Ω_r in the wedge left of the first slab, the right wall by the layers, which run from the slab boundary to the wall $X = w$.

Lemma 7.3 (end lemma). *Assume $g_0 > 1$, $t < 1$ and $m \tan \alpha + \sec \alpha \leq w$. Then the columns $K_0, \dots, K_{I-1}$, the layers Λ_j and the wedge rows Ω_r form a packing in Reg , and*

$$U \leq B^* := w \sin \alpha + w(1 - \cos \alpha) + \Delta(\sec \alpha + m \tan \alpha) + \lfloor \bar{g} \rfloor (1 + \tfrac{1}{2} \tan \alpha) + (1 + \mu \omega) \omega \\ + \lfloor g_0 \rfloor (1 + \tfrac{1}{2} \tan \alpha) + (\delta + 1) \tan \alpha (1 + \mu(\delta + 1) \tan \alpha), \quad \omega := \sec \alpha + \bar{g} \tan \alpha.$$

The seven terms are: the triangles beside the columns; the gaps above the columns (the staircase quantum $1 - \cos \alpha$); the slope of the ceiling across one column; the ends and fractional parts of the layers; the corner at the right wall; the wedge rows; and the corner at the left wall. No small-angle expansion is used.

Proof. (a) *The staircase exists up to the right wall.* For $0 \leq i < I$ we have $0 \leq \xi_i - m \tan \alpha = i \sec \alpha \leq \xi_i \leq w - \sec \alpha$, so $g_0 \leq \Gamma_i \leq \bar{g}$, and Γ_i is non-decreasing in i . H is affine in J with slope $1 - \cos \alpha > 0$ (note $t \geq t^* > 0$), so $J_i = \max\{J \in \mathbb{Z} : H(J) \leq \Gamma_i\}$. Moreover $J_i \geq 0$, because $H(0) = m \cos \alpha + \sin \alpha \leq g_0 \leq \Gamma_i$ by Lemma 7.2 ($t \geq t^*$ and $1 < g_0 < m$); $J_i \leq m - 1$, that is $n_i \geq 1$, because $H(m) = m + \sin \alpha > m > \bar{g} \geq \Gamma_i$; the gap $\Gamma_i - H(J_i)$ lies in $[0, 1 - \cos \alpha)$, because $H(J_i + 1) = H(J_i) + 1 - \cos \alpha > \Gamma_i$; and $J_0 \leq J_1 \leq \dots \leq J_{I-1}$. So every slab up to the right wall receives a column; the columns stop only because the width runs out, and $w - \xi_I < \sec \alpha$.

(b) *Columns lie in Reg.* By Lemma 7.1(iii),(iv) and $J_i + n_i = m$, every point of K_i has $X \geq \xi_i - m \tan \alpha \geq 0$ and $X \leq \xi_i + \sec \alpha \leq w$, and by (ii) and (a), $J_i \leq Y \leq H(J_i) \leq \Gamma_i = g(\xi_i - m \tan \alpha) \leq g(X)$.

(c) *Layers lie in Reg.* For $0 \leq j \leq \lfloor \bar{g} \rfloor$ we have $a_j \leq \xi_{p(j)} \leq \xi_I \leq w$ and $a_j \geq \xi_0 - j \tan \alpha = (m - j) \tan \alpha > 0$. If $j \leq j_1$, then $g(a_j) \geq j + 1$, so every point of Λ_j has $0 \leq a_j \leq X \leq a_j + L_j \leq w$ and $0 \leq Y \leq j + 1 \leq g(a_j) \leq g(X)$. Also $j_1 \leq \lfloor \bar{g} \rfloor - 1$: for $j = \lfloor \bar{g} \rfloor$, $g(a_j) \leq g(w) = \bar{g} < j + 1$.

(d) *Wedge rows lie in Reg.* For $r \leq \lfloor g_0 \rfloor - 1$, $m - r - 1 \geq m - \lfloor g_0 \rfloor > 0$, so $0 \leq l_r \leq (m - 1) \tan \alpha \leq w$, and $Y \leq r + 1 \leq \lfloor g_0 \rfloor \leq g_0 = g(0) \leq g(X)$ on Ω_r .

(e) *Disjointness.* Two closed convex sets on the two sides of a line have disjoint interiors; all separations below are of this kind.

- Columns: for $i < i'$, $K_i \subset \{\lambda \leq \xi_i + \sec \alpha\} = \{\lambda \leq \xi_{i+1}\} \subset \{\lambda \leq \xi_{i'}\}$ and $K_{i'} \subset \{\lambda \geq \xi_{i'}\}$ (Lemma 7.1(i)). No gap is needed between slabs, because the tilt never changes.
- Wedge rows against columns and layers: $\Omega_r \subset \{\lambda \leq l_r + (r + 1) \tan \alpha \leq (m - r - 1) \tan \alpha + (r + 1) \tan \alpha = \xi_0\}$, while the columns lie in $\{\lambda \geq \xi_0\}$ and $\Lambda_j \subset \{\lambda \geq a_j + j \tan \alpha = \xi_{p(j)} \geq \xi_0\}$ (as $X \geq a_j$, $Y \geq j$ and $\tan \alpha > 0$ on Λ_j).
- Layer Λ_j against column K_i : if $i < p(j)$, then $K_i \subset \{\lambda \leq \xi_{i+1}\} \subset \{\lambda \leq \xi_{p(j)}\}$ and $\Lambda_j \subset \{\lambda \geq \xi_{p(j)}\}$. If $i \geq p(j)$, then $J_i \geq j + 1$ (the J_i are non-decreasing, so $\{i : J_i \leq j\} = \{0, \dots, p(j) - 1\}$), hence $K_i \subset \{Y \geq j + 1\}$ and $\Lambda_j \subset \{Y \leq j + 1\}$.
- Layers among themselves, and wedge rows among themselves, lie in different rows $[j, j + 1]$.

(f) *A partition of Reg.* Put

$$\text{Wd} := \text{Reg} \cap \{\lambda < \xi_0\}, \quad \mathcal{S}_i := \text{Reg} \cap \{\xi_i \leq \lambda < \xi_{i+1}\} \quad (0 \leq i < I), \quad \text{Rt} := \text{Reg} \cap \{\lambda \geq \xi_I\},$$

$\mathcal{S}_i^+ := \mathcal{S}_i \cap \{Y \geq J_i\}$, $\mathcal{S}_i^- := \mathcal{S}_i \cap \{Y < J_i\}$ and $\text{Lay} := \bigcup_i \mathcal{S}_i^- \cup \text{Rt}$. Then Wd, the $\mathcal{S}_i^+$ and Lay partition Reg (as $\xi_{i+1} = \xi_i + \sec \alpha$, the slabs tile $[\xi_0, \xi_I]$). The interior of each placed piece lies in one part: $\text{int } K_i \subset \mathcal{S}_i^+$ by Lemma 7.1(i),(ii); $\text{int } \Omega_r \subset \text{Wd}$ by (e); and $\text{int } \Lambda_j \subset \text{Lay}$, because a point of Reg with $\lambda \geq \xi_{p(j)}$ and $Y < j + 1$ lies either in Rt or in some $\mathcal{S}_i$ with $i \geq p(j)$, where $J_i \geq j + 1 > Y$. Boundaries have measure 0 and the pieces have disjoint interiors, so U is the sum over the parts of the area of the part minus the number of unit squares in the pieces assigned to it.

(g) *The column parts.* Fix i and write $J = J_i$, $n = n_i$, $H_i = H(J)$. The parallelogram $\Pi_i := \{\xi_i \leq \lambda \leq \xi_i + \sec \alpha, J \leq Y \leq H_i\}$ lies in Reg: its points have $X \geq \xi_i - H_i \tan \alpha \geq \xi_i - m \tan \alpha \geq 0$ (as $H_i \leq \Gamma_i \leq \bar{g} < m$), $X \leq \xi_i + \sec \alpha \leq w$, and $Y \leq H_i \leq \Gamma_i = g(\xi_i - m \tan \alpha) \leq g(X)$. Its horizontal sections have length $\sec \alpha$, so $\text{area}(\Pi_i) = \sec \alpha(H_i - J) = \sec \alpha(n \cos \alpha + \sin \alpha) = n + \tan \alpha$. The rest of $\mathcal{S}_i^+$ lies in $\{Y > H_i\}$; there $X \leq \lambda < \xi_i + \sec \alpha$, so $Y \leq g(\xi_i + \sec \alpha) = \Gamma_i + \mu(\sec \alpha + m \tan \alpha)$, and the sections still have length at most $\sec \alpha$. Hence, by (a),

$$\begin{aligned} \text{area}(\mathcal{S}_i^+) - n_i &\leq \tan \alpha + \sec \alpha(\Gamma_i + \mu(\sec \alpha + m \tan \alpha) - H_i) \\ &< \tan \alpha + \sec \alpha(1 - \cos \alpha) + \sec \alpha \mu(\sec \alpha + m \tan \alpha). \end{aligned}$$

Summing, with $I \sec \alpha = \xi_I - \xi_0 \leq w$, $I \tan \alpha = I \sec \alpha \sin \alpha \leq w \sin \alpha$ and $w \mu = \Delta$,

$$\sum_i [\text{area}(\mathcal{S}_i^+) - n_i] \leq w \sin \alpha + w(1 - \cos \alpha) + \Delta(\sec \alpha + m \tan \alpha).$$

(h) *The layer part.* For an integer j let $\text{Lay}_j := \text{Lay} \cap \{j \leq Y < j + 1\}$. By the argument at the end of (f), Lay_j contains $\text{Reg} \cap \{j \leq Y < j + 1, \lambda \geq \xi_{p(j)}\}$; conversely, a point of $\mathcal{S}_i^-$ at height

$Y \in [j, j+1)$ has $J_i > Y \geq j$, hence $J_i \geq j+1$, that is $i \geq p(j)$ and $\lambda \geq \xi_{p(j)}$; and the points of Rt have $\lambda \geq \xi_I \geq \xi_{p(j)}$. So $\text{Lay}_j = \text{Reg} \cap \{j \leq Y < j+1, \lambda \geq \xi_{p(j)}\}$.

Rows $0 \leq j \leq j_1$. The section of Lay_j at height Y lies in $[\xi_{p(j)} - Y \tan \alpha, w]$, so $\text{area}(\text{Lay}_j) \leq \int_j^{j+1} (w - a_j + (Y - j) \tan \alpha) dY = w - a_j + \frac{1}{2} \tan \alpha$, while Λ_j covers $L_j > w - a_j - 1$ (if $L_j = 0$, nothing is placed and $w - a_j < 1$). So $\text{area}(\text{Lay}_j) - L_j < 1 + \frac{1}{2} \tan \alpha$, and there are $j_1 + 1 \leq \lfloor \bar{g} \rfloor$ such rows by (c).

The rest, $\text{Lay} \cap \{Y \geq j_1 + 1\}$. First, $j_1 \geq J_{I-1} - 1$: if $0 \leq j \leq J_{I-1} - 1$ (note $J_{I-1} - 1 \leq m - 2 = \lfloor \bar{g} \rfloor - 1$), then $p := p(j) \leq I - 1$ and $J_p \geq j + 1$, so $\Gamma_p \geq H(J_p) \geq J_p \geq j + 1$ (as $(m - J) \cos \alpha + \sin \alpha \geq 0$); and $a_j = \xi_p - j \tan \alpha \geq \xi_p - m \tan \alpha$, so $g(a_j) \geq \Gamma_p \geq j + 1$. Hence the points of $\mathcal{S}_i^-$, which have $Y < J_i \leq J_{I-1} \leq j_1 + 1$, are not in $\{Y \geq j_1 + 1\}$, and $\text{Lay} \cap \{Y \geq j_1 + 1\} = \text{Rt} \cap \{Y \geq j_1 + 1\}$. Next, $j_1 + 1 \leq \lfloor \bar{g} \rfloor$ by (c) and $j_1 + 1 \geq J_{I-1}$, so $p(j_1 + 1) = I$ and, by the maximality of j_1 , $g(a_{j_1+1}) < j_1 + 2$, where $a_{j_1+1} = \xi_I - (j_1 + 1) \tan \alpha \geq \xi_I - \bar{g} \tan \alpha > w - \sec \alpha - \bar{g} \tan \alpha = w - \omega$. Hence $\bar{g} - \mu\omega = g(w - \omega) < g(a_{j_1+1}) < j_1 + 2$, that is $\bar{g} - (j_1 + 1) < 1 + \mu\omega$. The sections of Rt at heights $Y \leq \bar{g}$ lie in $[\xi_I - Y \tan \alpha, w]$ and have length less than $\sec \alpha + \bar{g} \tan \alpha = \omega$, and Rt lies below $Y = \bar{g}$. So $\text{area}(\text{Rt} \cap \{Y \geq j_1 + 1\}) < \omega(1 + \mu\omega)$. In total,

$$\text{area}(\text{Lay}) - \sum_j L_j < \lfloor \bar{g} \rfloor (1 + \frac{1}{2} \tan \alpha) + (1 + \mu\omega) \omega.$$

(i) *The wedge part.* For $0 \leq r \leq \lfloor g_0 \rfloor - 1$, the section of Wd at height $Y \in [r, r+1)$ lies in $[0, \xi_0 - Y \tan \alpha]$ (note $\xi_0 - Y \tan \alpha = (m - Y) \tan \alpha > 0$), so $\text{area}(\text{Wd} \cap \{r \leq Y < r+1\}) \leq \xi_0 - (r + \frac{1}{2}) \tan \alpha$, while Ω_r covers $l_r > (m - r - 1) \tan \alpha - 1 = \xi_0 - (r + 1) \tan \alpha - 1$ (or nothing, when $(m - r - 1) \tan \alpha < 1$): the waste is less than $1 + \frac{1}{2} \tan \alpha$ per row, in $\lfloor g_0 \rfloor$ rows. The points of Wd with $Y \geq \lfloor g_0 \rfloor$ have $0 \leq X < \xi_0 - Y \tan \alpha \leq (m - \lfloor g_0 \rfloor) \tan \alpha < (\delta + 1) \tan \alpha$ and $Y \leq g(X) \leq g_0 + \mu(\delta + 1) \tan \alpha < \lfloor g_0 \rfloor + 1 + \mu(\delta + 1) \tan \alpha$, so their area is less than $(\delta + 1) \tan \alpha (1 + \mu(\delta + 1) \tan \alpha)$.

Adding (g), (h) and (i) gives $U \leq B^*$. $\square$

The bound holds for every y with $g_0 > 1$: no tuning of y , m or α is needed, and by (a) every slab up to the right wall gets a column.

7.2. The band ends.

Corollary 7.4. *Let $b \geq 10^4$ be an integer, $x := b^{-1/3}$, $\kappa := 3/4$, $\tan(\theta/2) = b/(b^2 - 1)$, and w, γ, σ as in Section 6. Let y be any real number with*

$$\kappa b^{2/3} \leq y \leq \kappa b^{2/3} + 2.$$

Apply the construction of Section 7.1 with $\sigma_0 = \sigma$ and $\mu = \tan \theta$ (so that $\text{Reg} = \text{Reg}(y)$ of Section 6), and with any integer $Q \geq 16b$. Then the construction gives a packing in $\text{Reg}(y)$ with

$$U \leq \Phi(x) b^{2/3} \leq \Phi(10^{-4/3}) b^{2/3} \leq 5.03 b^{2/3},$$

where Φ , defined in the proof, is non-decreasing on $(0, 10^{-4/3}]$, $\Phi(10^{-4/3}) \leq 5.0163$ and $\Phi(x) \rightarrow \sqrt{A/\kappa} + 3/2 \approx 4.3284$ as $x \rightarrow 0$, with $A := 6.000002$. Consequently the end cost of Proposition 6.3 satisfies $E(y) = U + \frac{1}{2} \tan \theta \leq 5.03 b^{2/3} + 1.0001/b$.

Proof. (i) *Band facts.* Write $t_\theta := b/(b^2 - 1)$ and $u := b^{1/3} = 1/x$. Then $t_\theta \leq 1.00000002/b$ (since $bt_\theta = b^2/(b^2 - 1)$), $\sigma < 2t_\theta \leq 2.0000001/b$ and $\mu = 2t_\theta/(1 - t_\theta^2) \leq 2.000001/b$; so $\Delta = w\mu < b\mu \leq 2.000001$ and $\sigma\mu \leq 4.1 \cdot 10^{-8}$. Also $w < b$ by Lemma 3.2, as $t_\theta > 1/b$, and $w \geq b\gamma = b(1 - 2t_\theta^2/(1 + t_\theta^2)) \geq b - 2bt_\theta^2 \geq b - 2.1/b$.

(ii) *Ranges.* $g_0 = y - \sigma\mu \in [\kappa u^2 - 4.1 \cdot 10^{-8}, \kappa u^2 + 2]$. Put $\nu := \kappa u^2 - 5 \cdot 10^{-8} = p(x)u^2$ with $p(x) := \kappa - 5 \cdot 10^{-8}x^2$; then $g_0 \geq \nu > 1$. Next $\bar{g} = y + (w - \sigma)\mu \leq y + \Delta \leq \kappa u^2 + 4.000001$, $m = \lfloor \bar{g} \rfloor + 1 \leq \kappa u^2 + 5.000001$, $\lfloor g_0 \rfloor \leq \kappa u^2 + 2$, and $\delta = m - g_0 \in (\Delta, \Delta + 1]$, so $\delta \leq \bar{\delta} := 3.000001$ and $\delta + 1 \leq 4.000001$.

(iii) *Tilt*. By Lemma 7.2 with $g_0 \geq \nu$, $\delta \leq \bar{\delta}$ and $A = 2\bar{\delta}$,

$$ut^* \leq u \frac{1 + \sqrt{1 + 2\nu\bar{\delta}}}{2\nu} = \frac{x + \sqrt{x^2 + Ap(x)}}{2p(x)} \leq \frac{x + \sqrt{x^2 + A\kappa}}{2p(x)},$$

and $u/Q \leq u/(16u^3) = x^2/16$. So $t \leq t^* + 1/Q \leq x\vartheta(x) =: \bar{t}(x)$ with

$$\vartheta(x) := \frac{x + \sqrt{x^2 + A\kappa}}{2p(x)} + \frac{x^2}{16}.$$

Put $\bar{\tau} := 2\bar{t}/(1 - \bar{t}^2)$ and $\bar{\varsigma} := (1 + \bar{t}^2)/(1 - \bar{t}^2)$. Then $\tan \alpha \leq \bar{\tau}$, $\sec \alpha \leq \bar{\varsigma}$, $\sin \alpha \leq 2t \leq 2\bar{t}$ and $1 - \cos \alpha \leq 2t^2 \leq 2\bar{t}^2$. At $x_0 := 10^{-4/3}$ (that is, $b = 10^4$), $\bar{t} \leq 0.0672$ and $\bar{\tau} \leq 0.1349$.

(iv) *The hypotheses of Lemma 7.3*. $g_0 > 1$ by (ii); $t \leq \bar{t}(x) \leq \bar{t}(x_0) \leq 0.0672 < 1$ by the monotonicity in (v); and $m \tan \alpha + \sec \alpha \leq (0.75b^{2/3} + 5.000001) \cdot 0.1349 + 1.01 < b - 2.1/b \leq w$, since $b^{2/3} \leq b/21.5$ for $b \geq 10^4$. Lemma 7.3 gives the packing and $U \leq B^*$. We divide each term of B^* by $u^2 = b^{2/3}$, using $w < b = u^3$, $\Delta \leq 2.000001$, $\mu \leq 2.000001 x^3$ and $\omega = \sec \alpha + \bar{g} \tan \alpha$:

$$\begin{aligned} T_1 &= w \sin \alpha / u^2 \leq 2ut \leq 2\vartheta(x), & T_2 &= w(1 - \cos \alpha) / u^2 \leq 2ut^2 \leq 2x\vartheta(x)^2, \\ T_3 &= \Delta(\sec \alpha + m \tan \alpha) / u^2 \leq 2.000001[x^2\bar{\varsigma} + (\kappa + 5.000001 x^2)\bar{\tau}], \\ T_4 &= [\bar{g}](1 + \tfrac{1}{2} \tan \alpha) / u^2 \leq (\kappa + 4.000001 x^2)(1 + \tfrac{1}{2} \bar{\tau}), \\ T_5 &= (1 + \mu\omega) \omega / u^2 \leq (1 + 2.000001 x \bar{\omega}) \bar{\omega}, & \bar{\omega} &:= x^2\bar{\varsigma} + (\kappa + 4.000001 x^2)\bar{\tau}, \\ T_6 &= [g_0](1 + \tfrac{1}{2} \tan \alpha) / u^2 \leq (\kappa + 2x^2)(1 + \tfrac{1}{2} \bar{\tau}), \\ T_7 &= (\delta + 1) \tan \alpha (1 + \mu(\delta + 1) \tan \alpha) / u^2 \leq 4.000001 \bar{\tau} x^2 (1 + 2.000001 \cdot 4.000001 x^3 \bar{\tau}). \end{aligned}$$

For T_5 we used $\omega/u^2 \leq \bar{\omega}$ and $\mu\omega \leq 2.000001 x^3 u^2 \bar{\omega} = 2.000001 x \bar{\omega}$. Let $\Phi(x)$ be the sum of the seven right-hand sides.

(v) *Monotonicity*. On $0 < x \leq x_0$, $p(x)$ is positive and decreasing and $x + \sqrt{x^2 + A\kappa}$ is increasing, so ϑ is increasing; $\bar{t} = x\vartheta$ is increasing and less than 1, so $\bar{\tau}$ and $\bar{\varsigma}$ are increasing. Every right-hand side in (iv) is a sum of products of non-negative non-decreasing functions of x . So Φ is non-decreasing in x , that is non-increasing in b , and $\Phi(x) \leq \Phi(x_0)$ for all $b \geq 10^4$.

(vi) *Values*. In interval arithmetic, $\Phi(x_0) \leq 5.01625$ (the terms are about 2.89126, 0.19400, 0.20946, 0.80975, 0.10545, 0.80515 and 0.00116). As $x \rightarrow 0$, $T_1 \rightarrow \sqrt{A\kappa}/\kappa = \sqrt{A/\kappa}$, $T_4, T_6 \rightarrow \kappa$ and the other terms tend to 0, so $\Phi \rightarrow \sqrt{A/\kappa} + 3/2$. Finally $A_0(y)$ exceeds the area of $\text{Reg}(y)$ by $\frac{1}{2} \tan \theta$ (the triangle with vertices $(0, y - \sigma\mu)$, (σ, y) , $(0, y + \gamma)$), and $\tan \theta \leq 2.000001/b$. $\square$

The values in (iii) and (vi) were certified in interval arithmetic (`mpmath.iv`, 40 digits, with x_0 enclosed in an interval) by the author's program and by three independent computations (two reviews and an independent reimplementation), which all give $\Phi(x_0) \in [5.0162487, 5.0162488]$; one of them also bounds Φ on all of $(0, x_0]$ by a subdivision into 4000 intervals, without using (v), and finds the maximum 5.0163 at x_0 . For larger b the constant improves: $\Phi(10^{-5/3}) \leq 4.6351$, $\Phi(10^{-2}) \leq 4.4685$, $\Phi(10^{-8/3}) \leq 4.3583$. The choice $\kappa = 3/4$ is close to the minimizer $(\sqrt{6}/4)^{2/3} \approx 0.721$ of the limit $\sqrt{6/\kappa} + 2\kappa$. The window has length $2 > \sec \theta$, which is what Section 7.3 needs: the bound holds for every y in the window, not only at sampled points.

7.3. Proof of Theorem 1.2. Let $C_E := 5.03$, $k \geq k_0 = 1.6 \cdot 10^7$, $b_* := (3k/(2C_E))^{3/5}$ (so $k = \frac{2}{3}C_E b_*^{5/3}$) and $b := \lceil b_* \rceil$. Since $k_0 \geq \frac{2}{3}C_E \cdot 10^{20/3} \approx 1.5565 \cdot 10^7$, we have $b \geq b_* \geq 10^4$; since $3/(2C_E) < 1$, $b \leq k^{3/5} + 1$. Take $\tan(\theta/2) = b/(b^2 - 1) > 1/b$; then $w < b$ and $S := k - b + w < k$. Use the packing of Corollary 6.2 with

$$y_0 := \tfrac{1}{4} \lceil 3b^{2/3} \rceil \in [\kappa b^{2/3}, \kappa b^{2/3} + \tfrac{1}{4}]$$

and, in each band, $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$. Here $h = b\sigma + \gamma < 3$ (as $b\sigma = 2b^2(b^2 - 1)/((b^2 - 1)^2 + b^2) < 2$) and $y_0 \leq 0.75(k^{3/5} + 1)^{2/3} + \frac{1}{4} \leq 0.75 k^{2/5} + 1$, so $h + 2y_0 < 1.5 k^{2/5} + 5 < k - b \leq L_b$;

hence $R \geq 1$ and $y_1 = L_b - h - (R - 1)d - y_0 \in [y_0, y_0 + d]$. As $d < 1.0001$, both y_0 and y_1 lie in the window of Corollary 7.4, whatever value y_1 takes. Fill the four end regions $\text{Reg}(y_0)$, $\text{Reg}(y_1)$ of the two bands by Corollary 7.4 with Q the least power of 2 that is at least $16b$. By Corollary 6.2 this is a packing in $[0, S]^2$, and by Proposition 6.3

$$k^2 - N = (k^2 - S^2) + \sum_{2 \text{ bands}} [(R - 1) \tan \theta + E(y_0) + E(y_1)].$$

- $(R - 1) \tan \theta = (R - 1)d \sin \theta = (L_b - h - y_0 - y_1) \sin \theta \leq L_b \sigma < 2L_b t_\theta$, so the two bands give less than $(S + k - b) 2t_\theta < 4kb/(b^2 - 1) = 4k/b + 4k/(b(b^2 - 1))$.
- $b - w = 2t_\theta(bt_\theta - 1)/(1 + t_\theta^2) = 2t_\theta/((b^2 - 1)(1 + t_\theta^2)) < 2b/(b^2 - 1)^2$, so $k^2 - S^2 < 2k(b - w) < 4kb/(b^2 - 1)^2$.
- $\sum E \leq 4(C_E b^{2/3} + \frac{1}{2} \tan \theta)$ by Corollary 7.4.

Hence $k^2 - N < 4k/b + 4C_E b^{2/3} + \lambda'$ with $\lambda' := 4k/(b(b^2 - 1)) + 4kb/(b^2 - 1)^2 + 2 \tan \theta \leq 8.001 k/b^3 + 4.1/b$. Since $k \leq \frac{2}{3} C_E b^{5/3}$, $8.001 k/b^3 \leq 8.001 \cdot \frac{2}{3} C_E b^{-4/3} \leq 1.25 \cdot 10^{-4}$, and $4.1/b \leq 4.1 \cdot 10^{-4}$; so $\lambda' < 1$, and Lemma 2.2 gives $c^*(k) \leq k^2 - N - 1 < 4k/b + 4C_E b^{2/3}$. Finally $4k/b \leq 4k/b_* = \frac{8}{3} C_E b_*^{2/3}$ and $b^{2/3} \leq (b_* + 1)^{2/3} \leq b_*^{2/3} + \frac{2}{3} b_*^{-1/3} \leq b_*^{2/3} + \frac{2}{3} \cdot 10^{-4/3}$, so

$$c^*(k) < \frac{20}{3} C_E b_*^{2/3} + \frac{8}{3} C_E \cdot 10^{-4/3} = \frac{20}{3} \left(\frac{3}{2}\right)^{2/5} C_E^{3/5} k^{2/5} + \frac{8}{3} C_E \cdot 10^{-4/3}.$$

In interval arithmetic,

$$\frac{20}{3} \left(\frac{3}{2}\right)^{2/5} 5.03^{3/5} \in [20.66740, 20.66741] \quad \text{and} \quad \frac{8}{3} \cdot 5.03 \cdot 10^{-4/3} \in [0.62259, 0.62260],$$

which gives Theorem 1.2. For every $C > \sqrt{A/\kappa} + 3/2 \approx 4.3284$ we have $\Phi(b^{-1/3}) \leq C$ for all large b , and the same argument with C in place of C_E gives $c^*(k) \leq (\frac{20}{3} (\frac{3}{2})^{2/5} C^{3/5} + o(1)) k^{2/5}$; letting C tend to the limit gives $c^*(k) \leq (18.89 + o(1)) k^{2/5}$. $\square$

Remark 7.5 (comparison and exact checks). The construction of Section 7.1 was implemented in exact rational arithmetic, with Q the least power of 2 that is at least $16b$. For $b = 400, 1600, 6400, 25600, 102400$ and five lifts each in the window of Corollary 7.4, all 25 end packings were checked by an exact checker that does not use Lemma 7.3 (shapes, containment, separating axis tests for all pairs of pieces with overlapping bounding boxes; for $b = 102400$ also by a check of the separating lines of the proof); all pass, $U \leq B^*$ holds in each, and $U/b^{2/3}$ stays between 3.26 and 4.36. Deliberately corrupted versions (a column shifted by 2^{-30} , a layer lengthened, a column given one more square, and so on) were rejected. The full packings of Section 7.3, counted exactly (the four ends checked by exact containment and by the separating lines of the proof; the rows not checked, their correctness rests on Lemmas 3.3 and 6.1; the identity of Proposition 6.3 checked exactly), give:

k	b	y_0	$c^*(k) \leq$	$20.668 k^{2/5} + 0.623$
10^5	484	185/4	1794	$(k < k_0)$
10^6	1927	465/4	4467	$(k < k_0)$
10^7	7669	1167/4	10579	$(k < k_0)$
$1.6 \cdot 10^7$	10167	352	12707	15738.5
$3.825 \cdot 10^7$	17152	499	17865	22303.0
10^8	30530	2931/4	26998	32757.2

For $k < k_0$ these packings are still valid (Lemma 7.3 does not need $b \geq 10^4$); only the closed form of Theorem 1.2 is restricted to $k \geq k_0$. An independent reimplementaion, written from the text of the proof only (with its own admissible choices of Q), checked the end packings exactly for b up to $4 \cdot 10^5$ and many lifts in the window, and the full packings for $k = 1.6 \cdot 10^7, 3.825 \cdot 10^7, 10^8, 10^9$ and $3 \cdot 10^9$: no overlaps, no containment errors, $U \leq B^*$ in every case, and $c^*(10^9) \leq 67065$,

$c^*(3 \cdot 10^9) \leq 99640$ (against 82281.4 and 127687.7 from Theorem 1.2); 8 of 8 deliberately corrupted versions were rejected. It relies on Lemmas 3.3 and 6.1 for the rows. The tuned packings of Theorem 1.6 are 4.9–11.8% better than these (Section 9).

8. CHAINED WALL FILLERS AND THE EXPONENT $3/8$

In the end lemma of Section 7 the columns cost about $b/\sqrt{m}$ and the two walls about m each. In this section the walls are filled more efficiently: each wall is a thin trapezoid, and it is filled by a chain of staircases (*blocks*) whose tilt is lowered slightly from block to block, so that the horizontal layers opened in one block close just before the next block. The wall cost becomes $O(m^{3/4})$, the best column length is $m \asymp b^{4/5}$, the end cost is $O(b^{3/5})$, and Proposition 6.3 gives Theorem 1.1. The ideas (horizontal layers under tilted columns, and filling the wall trapezoids by stairs whose tilt is lowered from block to block) are those of Bui [Bui, Sections 3–4]; the construction and the proof below are self-contained and do not use any argument of [Bui].

Throughout this section we use the tilt notation of Section 7.1, the columns $K(\xi, J, n)$ and Lemmas 7.1 and 7.2, and we put

$$q := 1 - \cos \alpha = \frac{\tan \alpha^2}{\sec \alpha (\sec \alpha + 1)} \leq \frac{\tan \alpha^2}{2}, \quad \text{so that} \quad \sec \alpha - 1 = q \sec \alpha, \quad \sec \alpha \sin \alpha = \tan \alpha.$$

q is increasing in t . On $t \in (0, 1/10]$ we have $\tan \alpha \leq 2.1t$, and $t \mapsto \tan \alpha$ has derivative $2(1+t^2)/(1-t^2)^2 \leq 2.1$, so neighbouring points of a grid $(1/Q^*)\mathbb{Z}$ differ in $\tan \alpha$ by at most $\delta_Q := 2.1/Q^*$. A tilt with index 0 (as t_0, α_0) is the *initial tilt* of a wall. Two closed convex sets that lie in the two closed half-planes of one line have disjoint interiors; every disjointness statement below is of this kind.

Grids. Q is the grid of the main tilt: any power of 2 with $Q \geq 16b$. The wall tilts and the cut tilt use the grid $Q^* := \max(2^{40}, Q)$, and the phases the grid $Q_f := \max(2^{48}, Q)$.

8.1. The wall filler. For reals $\chi \geq 4$, $\zeta > 0$ and $L \geq 0$ let $r(x) := \chi + \zeta x$ and

$$Z = Z(\chi, \zeta, L) := \{(x, y) : 0 \leq x \leq L, 0 \leq y \leq r(x)\},$$

a right trapezoid of area $L\chi + \zeta L^2/2$. The input is a grid tilt $t_0 \in (0, 1/10] \cap (1/Q^*)\mathbb{Z}$; put $N_0 := \lfloor \chi \rfloor - 1 \geq 3$. For a block β with tilt t_β we write $\lambda_\beta(x, y) := x + y \tan \alpha_\beta$.

Construction ZC.

(Z1) *Block 0:* tilt t_0 , base $F_0 := \lfloor \chi - N_0 \cos \alpha_0 - \sin \alpha_0 \rfloor \geq 0$, total length $M_0 := F_0 + N_0$, offset $s_0 := M_0 \tan \alpha_0$.

(Z2) *Block β* has tilt t_β (with $\cos \alpha, \sin \alpha, \tan \alpha, \sec \alpha, q$), offset s_β , base F_β and total length $M_\beta = F_\beta + N_0$. If $s_\beta + \sec \alpha > L$ the chain stops (block β does not exist). Otherwise, for $i = 0, 1, 2, \dots$ let $\xi_i := s_\beta + i \sec \alpha$; stop (reason ‘ L ’) if $\xi_i + \sec \alpha > L$; else let $\Gamma_i := r(\xi_i - M_\beta \tan \alpha)$ and $\tilde{J}_i := \lfloor (\Gamma_i - M_\beta \cos \alpha - \sin \alpha)/q \rfloor$; stop (reason ‘saturation’) if $\tilde{J}_i > M_\beta - 1$; else place the column $K(\xi_i, J_i, n_i)$ with $J_i := \tilde{J}_i$, $n_i := M_\beta - J_i$. Let P_β be the number of columns and $\hat{\xi}_\beta := s_\beta + P_\beta \sec \alpha$.

(Z3) *Next block.* If block β stopped by saturation (it is *complete*), let $\kappa_\beta := q^2/\zeta$, let $t_{\beta+1}$ be the largest grid point $t' \leq t_\beta$ with $\tan \alpha' \leq \tan \alpha - \kappa_\beta$, and put $\hat{\kappa}_\beta := \tan \alpha - \tan \alpha_{\beta+1} \geq \kappa_\beta$ and $\eta_\beta := \hat{\kappa}_\beta - \kappa_\beta \in [0, \delta_Q)$. *Phase:* let $J_{l_0} := J_0$ and $J_L := J_{P_\beta-1}$ (indices of block β). If $J_L > J_{l_0}$, put $A := (\Gamma_0 - M_\beta \cos \alpha - \sin \alpha)/q$, $B := q/(\zeta \sec \alpha)$, $p := \#\{i : J_i \leq J_{l_0}\}$, $f := p - B(J_{l_0} + 1 - A)$,

$$\psi := \hat{\xi}_\beta - \tan \alpha_{\beta+1} - (s_\beta + p \sec \alpha) + J_{l_0} \hat{\kappa}_\beta, \quad \omega_0 := \text{frac}(-(\psi + f(\sec \alpha - 1) - (\sec \alpha - 1))),$$

and $\omega := \lceil \omega_0 Q_f \rceil / Q_f \in [0, 1]$; otherwise $\omega := 0$. Then $s_{\beta+1} := \hat{\xi}_\beta + \omega$ and, with primes for the tilt $t_{\beta+1}$,

$$\begin{aligned} F_{\beta+1} &:= \max\{J \in \mathbb{Z} : J + N_0 \cos \alpha' + \sin \alpha' \leq r(s_{\beta+1} - (J + N_0) \tan \alpha')\} \\ &= \left\lfloor \frac{\chi + \zeta s_{\beta+1} - N_0(\cos \alpha' + \zeta \tan \alpha') - \sin \alpha'}{1 + \zeta \tan \alpha'} \right\rfloor, \end{aligned}$$

$$M_{\beta+1} := F_{\beta+1} + N_0.$$

(Z4) *Rows.* Block β blocks row j if $J_{\beta,0} \leq j$ ($J_{\beta,i}$ is J_i of block β). For each integer $0 \leq j < \lfloor r(L) \rfloor$, walk from $x = 0$ to $x = L$: a *stretch* starts at $x = 0$ or just after a blocking block β , at the abscissa $\xi_{\beta,p} - j \tan \alpha_\beta$ with $p := \#\{i : J_{\beta,i} \leq j\}$ ($p = P_\beta$ allowed, $\xi_{\beta,P_\beta} = \hat{\xi}_\beta$), and ends at the next blocking block β_2 , at the abscissa $s_{\beta_2} - (j+1) \tan \alpha_{\beta_2}$, or at $x = L$. For a stretch from a to a^+ with $a^+ - a \geq 1$, $a \geq 0$ and $j+1 \leq r(a)$, place the row $[a, a + \lfloor a^+ - a \rfloor] \times [j, j+1]$.

Each block is one staircase of Section 7.1 (one tilt, total length M_β , J non-decreasing); a block ends when its staircase is used up, and the next block starts after a phase gap $\omega < 1$ with the slightly smaller tilt, chosen so that every layer opened in the block ends just before the next block with a small residual (see (R) below).

Parameters. Let $\varpi \in (0, 1)$ (we use $\varpi = 0.65$), let $q_{\min}$ be the value of q at the slope $\varpi \tan \alpha_0$, that is $q_{\min} := (\varpi \tan \alpha_0)^2 / (s_m(s_m + 1))$ with $s_m := \sqrt{1 + \varpi^2 \tan^2 \alpha_0}$, and put $M_{\max} := r(L) + N_0 q_0$ (q_0 is q at t_0),

$$\Lambda := \frac{q_{\min} N_0 - 1 - \zeta \tan \alpha_0}{\zeta}, \quad \bar{K} := \frac{L}{\Lambda}, \quad \varepsilon_W := (\sec \alpha_0 - 1) + N_0 \delta_Q + \frac{1}{Q_f}.$$

Hypotheses.

(W1) $\Lambda \geq N_0 \tan \alpha_0$ (in particular $\Lambda > 0$);

(W2) $\Delta_{\text{dr}} := \frac{q_0}{N_0} \left(L + \bar{K} \frac{1 + \zeta \tan \alpha_0}{\zeta} \right) + \bar{K} \delta_Q < (1 - \varpi) \tan \alpha_0$;

(W3) $N_0 \cos \alpha_0 \geq \frac{\zeta \sec \alpha_0}{q_{\min}} + \tan \alpha_0 + \zeta(2 + \sec \alpha_0 + M_{\max} \tan \alpha_0) + \frac{1 + \zeta \tan \alpha_0}{q_{\min}}$;

(W4) $\varepsilon_W < 1$;

(W5) $\delta_Q \leq \varpi \tan \alpha_0$ and $t_0 \leq 1/10$.

Lemma 8.1 (wall filler). *Under (W1)–(W5) the construction ZC does not fail, its pieces form a packing in Z , and the uncovered area $U_Z := \text{area}(Z) - \# \text{squares}$ satisfies*

$$\begin{aligned} U_Z \leq B_W &:= L c_{\text{col}} + \bar{K} \left[(N_0 - 1)(\varepsilon_W + \tan \alpha_0) + \frac{1 + \zeta \tan \alpha_0}{q_{\min}} (1 + \tan \alpha_0) + A_T \right] \\ &\quad + \lfloor \chi \rfloor (1 + \tfrac{1}{2} \tan \alpha_0) + w_0 (1 + \zeta w_0) + r(L) (1 + \tfrac{1}{2} \tan \alpha_0) + w_e (4 + 2\zeta w_e), \end{aligned}$$

where $c_{\text{col}} := \tan \alpha_0 + \sec \alpha_0 q_0 + \sec \alpha_0 \zeta (\sec \alpha_0 + N_0 q_0 \tan \alpha_0)$, $A_T := h_T w_T$,

$$h_T := 1 + \frac{\zeta \sec \alpha_0}{q_{\min}} + \tan \alpha_0 + \zeta(1 + \sec \alpha_0 + M_{\max} \tan \alpha_0), \quad w_T := 1 + (r(L) + 1) \left(\frac{q_0^2}{\zeta} + \delta_Q \right),$$

$w_0 := (1 + N_0 q_0) \tan \alpha_0$ and $w_e := 1 + \sec \alpha_0 + (r(L) + N_0 q_0) \tan \alpha_0$.

The terms are: column triangles, gaps and ceiling slope; aligned closings; unaligned closings; transition slivers; the left end; the right end.

Proof. (a) *One block.* Fix an existing block β with tilt t ($\cos \alpha, \sin \alpha, \tan \alpha, \sec \alpha, q$), offset s , base F and total $M = F + N_0$, and assume the invariants

(I1) $\tan \alpha \geq \varpi \tan \alpha_0$ (hence $q \geq q_{\min}$);

(I2) $s \geq M \tan \alpha$;

(I3) $F \geq 0$ and $F + N_0 \cos \alpha + \sin \alpha \leq r(s - M \tan \alpha) < F + 1 + N_0 \cos \alpha + \sin \alpha + \zeta \tan \alpha$.

(I3) holds for $\beta = 0$ ($\Gamma_0 = r(0) = \chi$ and the definition of F_0) and for $\beta + 1$ by the definition of $F_{\beta+1}$ as a maximum ($J = F_{\beta+1} + 1$ violates the defining inequality, and $r(s' - (F' + 1 + N_0) \tan \alpha') = r(s' - M' \tan \alpha') - \zeta \tan \alpha'$). $\Gamma_i = r(s - M \tan \alpha) + i\zeta \sec \alpha$ is affine in i , so $\tilde{J}_i = \lfloor A + i/B \rfloor$, and the J_i are non-decreasing. $H(J) := J + (M - J) \cos \alpha + \sin \alpha$ is affine in J with slope q , so $J_i = \max\{J : H(J) \leq \Gamma_i\}$ and the gap $\Gamma_i - H(J_i)$ lies in $[0, q]$. $J_0 \geq F$ because $H(F) = F + N_0 \cos \alpha + \sin \alpha \leq \Gamma_0$ by (I3); so $J_i \geq F$, $n_i \leq N_0$, and $n_i \geq 1$ (no saturation). By (I3), $A < F + (1 + \zeta \tan \alpha)/q$, so $J_0 < F + (1 + \zeta \tan \alpha)/q$. The block is not empty: it exists, so $s + \sec \alpha \leq L$, and $J_0 < F + (1 + \zeta \tan \alpha_0)/q_{\min} < F + N_0$ by (W1); so $\tilde{J}_0 \leq M - 1$ and $P \geq 1$. *Containment of columns* (Lemma 7.1, $J_i + n_i = M$): $X \geq \xi_i - M \tan \alpha \geq s - M \tan \alpha \geq 0$ by (I2), $X \leq \xi_i + \sec \alpha \leq L$, and $J_i \leq Y \leq H(J_i) \leq \Gamma_i = r(\xi_i - M \tan \alpha) \leq r(X)$. *Length of a complete block*. Saturation at $i = P$ means $A + P/B \geq M$, so $P > B(N_0 - (1 + \zeta \tan \alpha)/q)$ and

$$(8.1) \quad \Lambda_\beta := P \sec \alpha > \frac{qN_0 - 1 - \zeta \tan \alpha}{\zeta} \geq \Lambda, \quad \text{equivalently} \quad q \leq \frac{\zeta \Lambda_\beta + 1 + \zeta \tan \alpha}{N_0}$$

(the middle expression is increasing in q and decreasing in $\tan \alpha$; use $q \geq q_{\min}$, $\tan \alpha \leq \tan \alpha_0$). *Last column of a complete block*: $J_L := J_{P-1} \geq \tilde{J}_P - 1/B - 1 \geq M - 1/B - 1$, so $n_L := M - J_L \leq 1 + 1/B = 1 + \zeta \sec \alpha/q$.

(b) *Passing to block $\beta + 1$* (β complete; primes refer to $t_{\beta+1}$). *Tilt drift*. Let C_β be the sum of $\kappa_\gamma + \delta_Q$ over the complete blocks $\gamma \leq \beta$, and suppose (I1) for the blocks $0, \dots, \beta$. By (8.1) these blocks have total length at least $(\beta + 1)\Lambda$ inside $[0, L]$, so $\beta + 1 \leq \bar{K}$; and by (8.1), $\kappa_\gamma = q_\gamma^2/\zeta \leq (q_0/N_0)(\Lambda_\gamma + (1 + \zeta \tan \alpha_0)/\zeta)$. Summing ($\sum \Lambda_\gamma \leq L$) gives $C_\beta \leq \Delta_{\text{dr}} < (1 - \varpi) \tan \alpha_0$ by (W2). By induction $\tan \alpha_\beta \geq \tan \alpha_0 - C_{\beta-1}$, so $\tan \alpha - \kappa_\beta \geq \tan \alpha_0 - C_\beta + \delta_Q > \varpi \tan \alpha_0 + \delta_Q \geq \tan \alpha(1/Q^*)$ by (W5): a positive grid point t' exists, and $\tan \alpha' > \tan \alpha - \kappa_\beta - \delta_Q \geq \tan \alpha_0 - C_\beta > \varpi \tan \alpha_0$. So (I1) holds for $\beta + 1$, and $t' \leq t$, $\cos \alpha' \geq \cos \alpha$. $F' \geq F$. $J = F$ is admissible in the definition of F' : $r(s' - M \tan \alpha') \geq r(\hat{\xi} - M \tan \alpha) = \Gamma_P$, saturation gives $\Gamma_P \geq M \cos \alpha + \sin \alpha + qM = M + \sin \alpha$, and $F + N_0 \cos \alpha' + \sin \alpha' \leq M + \sin \alpha' \leq M + \sin \alpha$. $J'_0 \leq J_L$. By maximality of J_L , $\Gamma_{P-1} < H(J_L + 1) \leq J_L + n_L + \sin \alpha$. As $s' \leq \hat{\xi} + 1 = \xi_{P-1} + \sec \alpha + 1$, $r(s') \leq \Gamma_{P-1} + \zeta(1 + \sec \alpha + M \tan \alpha)$. By (I3) for $\beta + 1$, $F' + N_0 \cos \alpha' + \sin \alpha' \leq r(s' - M' \tan \alpha') \leq r(s')$, so $F' < J_L + n_L + \sin \alpha - N_0 \cos \alpha' + \zeta(1 + \sec \alpha + M \tan \alpha)$; and by (a) for block $\beta + 1$, $J'_0 < F' + (1 + \zeta \tan \alpha')/q'$. With $n_L \leq 1 + \zeta \sec \alpha/q$, $M \leq M_{\max}$ (from $F + N_0 \cos \alpha + \sin \alpha \leq r(L)$), $\cos \alpha' \geq \cos \alpha_0$, $q, q' \geq q_{\min}$, $\sec \alpha \leq \sec \alpha_0$, $\sin \alpha \leq \tan \alpha_0$, hypothesis (W3) gives $J'_0 < J_L + 1$. (I2) for $\beta + 1$: $M' = F' + N_0 \leq J'_0 + N_0 \leq J_L + N_0 \leq M - 1 + N_0$, so $M' \tan \alpha' \leq M \tan \alpha + N_0 \tan \alpha_0 \leq s + \Lambda_\beta = \hat{\xi} \leq s'$ by (I2), (8.1) and (W1). *Aligned closings*. For $J_{l_0} \leq j < J_L$ put $p(j) := \#\{i : J_i \leq j\}$; then $1 \leq p(j) \leq P - 1$ and, since $j \geq \lfloor A \rfloor = J_{l_0}$, $p(j) = \lceil B(j+1-A) \rceil = B(j+1-A) + f_j$ with $f_j \in [0, 1)$. The stretch of row j from $a_j = \xi_{p(j)} - j \tan \alpha$ to $s' - (j+1) \tan \alpha'$ has length $\ell_j = s' - \tan \alpha' - s - p(j) \sec \alpha + j\hat{\kappa}$. Modulo 1 (as $p(j)$ is an integer and $\sec \alpha = 1 + (\sec \alpha - 1)$, and $B(\sec \alpha - 1) = q(\sec \alpha - 1)/(\zeta \sec \alpha) = q^2/\zeta = \kappa$),

$$\ell_j \equiv \omega + C + j\eta - f_j(\sec \alpha - 1), \quad C := \hat{\xi} - \tan \alpha' - s - B(1 - A)(\sec \alpha - 1).$$

By (Z3), $\psi + f(\sec \alpha - 1) \equiv C + J_{l_0}\eta$, so $\omega_0 + C + J_{l_0}\eta - (\sec \alpha - 1)$ is an integer and

$$\ell_j \equiv (\omega - \omega_0) + (j - J_{l_0})\eta + (1 - f_j)(\sec \alpha - 1) \pmod{1},$$

where $0 \leq \omega - \omega_0 < 1/Q_f$, $0 \leq (j - J_{l_0})\eta < N_0\delta_Q$ (as $J_L - J_{l_0} \leq N_0 - 1$) and $0 < (1 - f_j)(\sec \alpha - 1) \leq \sec \alpha_0 - 1$. The right-hand side lies in $(0, \varepsilon_W]$ with $\varepsilon_W < 1$ (W4). Also $\ell_j \geq \hat{\xi} - \xi_{p(j)} - \tan \alpha' \geq \sec \alpha - \tan \alpha' > 0$. Hence

$$(R) \quad \ell_j \geq 0 \quad \text{and} \quad \text{frac}(\ell_j) \leq \varepsilon_W.$$

(c) *Global structure*. By (a), (b) and induction every block satisfies (I1)–(I3), ZC never fails, the bases F_β are non-decreasing, the tilts non-increasing, $s_{\beta+1} \geq \hat{\xi}_\beta$, $J_{\beta+1,0} \leq J_{\beta,L}$, and there are at most $\bar{K}$ complete blocks. For $Y \geq 0$ and $\beta < \beta''$ we have $\lambda_{\beta''} \leq \lambda_\beta$. A stretch of row j runs from just after a blocking block β (left line $\lambda_\beta = \xi_{\beta,p(j)}$) to the next blocking block $\beta_2 > \beta$ (line

$\lambda_{\beta_2} = s_{\beta_2}$), passing below all blocks in between, whose columns have $J \geq J_{\beta'',0} \geq j+1$. If β is not the last block and $\beta+1$ does not block row j , then $j < J_{\beta+1,0} \leq J_{\beta,L}$, so $p(j) \leq P_\beta - 1$. Hence $p(j) = P_\beta$ (left end on $\lambda_\beta = \hat{\xi}_\beta$) occurs only for stretches that end at the next block's line (*transition rows*, $j \geq J_{\beta,L}$) or for the last block (*end rows*).

(d) *Containment of rows*. A placed row has $a \geq 0$, $a + \lfloor a^+ - a \rfloor \leq a^+ \leq L$ (as a^+ is L or $s_\beta - (j+1)\tan\alpha_\beta \leq s_\beta \leq L$), and $j+1 \leq r(a) \leq r(X)$ on it. So every piece lies in Z .

(e) *Disjointness*. Columns of one block lie in the slabs $\xi_i \leq \lambda_\beta \leq \xi_{i+1}$ (Lemma 7.1(i)). For $\beta < \beta''$, a column of β lies in $\{\lambda_\beta \leq \hat{\xi}_\beta\} \subset \{\lambda_{\beta''} \leq \hat{\xi}_{\beta''}\} \subset \{\lambda_{\beta''} \leq s_{\beta''}\}$ (as $Y \geq 0$), and a column of β'' in $\{\lambda_{\beta''} \geq s_{\beta''}\}$. A row piece of a stretch of row j from block β to block β_2 (or $x = L$): all columns of blocks $\gamma < \beta$ and the columns $i < p(j)$ of β lie in $\{\lambda_\beta \leq \xi_{\beta,p(j)}\}$, and the row piece in $\{\lambda_\beta \geq \xi_{\beta,p(j)}\}$ (its lower left corner is on that line and $\tan\alpha_\beta > 0$); the columns $i \geq p(j)$ of β and all columns of the blocks strictly between β and β_2 have $J \geq j+1$, so they lie in $\{Y \geq j+1\}$ and the row piece in $\{Y \leq j+1\}$; the columns of β_2 and of later blocks lie in $\{\lambda_{\beta_2} \geq s_{\beta_2}\}$, and the row piece in $\{\lambda_{\beta_2} \leq s_{\beta_2}\}$ (its top right corner is on or left of that line). For the first stretch of a row (left end $x = 0$) only the last two cases occur. Different rows are separated by the lines $Y = j+1$, and the stretches of one row by the lines above. So ZC is a packing in Z .

(f) *Accounting*. For each column $K_{\beta,i}$ let its *cell* be $\{z \in Z : \xi_i \leq \lambda_\beta < \xi_{i+1}, Y \geq J_i\}$. The cells are pairwise disjoint: slabs of one block are disjoint, and with $E_\beta := \{z \in Z : \lambda_\beta(z) < s_\beta\}$ we have $E_0 \subset E_1 \subset \dots$, and the cells of block β lie in $E_{\beta+1} \setminus E_\beta$. For $Y \in [j, j+1)$ and a slab of a blocking block, $Y \geq J_i$ iff $J_i \leq j$; so the complement of the cells in $Z \cap \{j \leq Y < j+1\}$ is the disjoint union of the *stretch regions* $\{j \leq Y < j+1, \lambda_\beta \geq \xi_{\beta,p(j)} \text{ (or } X \geq 0), \lambda_{\beta_2} < s_{\beta_2} \text{ (or } X \leq L)\} \cap Z$. The interior of every placed piece lies in its cell or stretch region, and boundaries have measure 0; so U_Z is the sum over the cells of (area $- n$) plus the sum over the stretch regions of (area $-$ placed squares).

(f1) *Cells*. The parallelogram $\{\xi_i \leq \lambda_\beta \leq \xi_i + \sec\alpha, J_i \leq Y \leq H(J_i)\}$ has area $\sec\alpha(n \cos\alpha + \sin\alpha) = n + \tan\alpha$, so the part of the cell in $\{Y \leq H(J_i)\}$ has area at most $n + \tan\alpha$. The rest of the cell lies in $\{Y > H(J_i)\}$, where $X \leq \xi_i + \sec\alpha - H(J_i)\tan\alpha$, so $Y \leq r(\xi_i + \sec\alpha - H(J_i)\tan\alpha) = \Gamma_i + \zeta(\sec\alpha + (M - H(J_i))\tan\alpha)$ with $M - H(J_i) = nq - \sin\alpha \leq N_0q$; its sections have length at most $\sec\alpha$. So (area of the cell) $- n \leq \tan\alpha + \sec\alpha q + \sec\alpha \zeta(\sec\alpha + N_0q \tan\alpha) \leq c_{\text{col}}$ (all terms increase with $t \leq t_0$, and the gap is less than q). The slab of column (β, i) meets the line $Y = 0$ in $[\xi_i, \xi_i + \sec\alpha] \subset [0, L]$; these intervals have disjoint interiors (consecutive inside a block, and $s_{\beta+1} \geq \hat{\xi}_\beta$) and length $\sec\alpha \geq 1$, so there are at most L columns, and the cells cost at most Lc_{col} .

(f2) *Stretch areas*. A stretch region between the lines $\lambda_\beta = c$ and $\lambda_{\beta_2} = s'$ has horizontal sections of length $(s' - Y \tan\alpha_{\beta_2}) - (c - Y \tan\alpha_\beta)$, affine in Y with slope $\tan\alpha_\beta - \tan\alpha_{\beta_2} \geq 0$; so its area is at most $\ell + \tan\alpha_0$, where $\ell := s' - (j+1)\tan\alpha_{\beta_2} - (c - j \tan\alpha_\beta)$ is the stretch length of (Z4) (with $\frac{1}{2}\tan\alpha_0$ for an end at $x = 0$ or $x = L$). A placed piece leaves at most $\text{frac}(\ell) + \tan\alpha_0$; if $0 \leq \ell < 1$ the waste is at most $\ell + \tan\alpha_0$ as well.

(f3) *The stretches*, each counted by its right end.

- Right end at $s_{\beta+1}$, left end in block β with $J_{\beta,0} \leq j < J_{\beta,L}$ (*aligned closings*): by (R), $\ell \geq 0$ and $\text{frac}(\ell) \leq \varepsilon_W$; the left end has $a_j \geq \xi_p - M \tan\alpha \geq 0$ and $j+1 \leq J_{p(j)} \leq H(J_{p(j)}) \leq \Gamma_{p(j)} \leq r(a_j)$, so the piece is placed when $\ell \geq 1$. Waste at most $\varepsilon_W + \tan\alpha_0$, in at most $N_0 - 1$ rows per complete block.
- Right end at $s_{\beta+1}$, left end before block β ($j < J_{\beta,0}$; *unaligned closings*): these rows have $J_{\beta+1,0} \leq j < J_{\beta,0}$, at most $J_{\beta,0} - F_\beta < (1 + \zeta \tan\alpha_0)/q_{\min}$ per complete block; the left end is $x = 0$ or a slab line with $p \leq P - 1$, and the waste is at most $1 + \tan\alpha_0$.
- Right end at $s_{\beta+1}$, left end on $\hat{\xi}_\beta$ ($j \geq J_{\beta,L}$; *transition rows*): their union lies in $\{\lambda_\beta \geq \hat{\xi}_\beta, \lambda_{\beta+1} < s_{\beta+1}, Y \geq J_{\beta,L}\}$. Its points have $X < s_{\beta+1}$, so $Y \leq r(s_{\beta+1}) < J_L + n_L + \sin\alpha +$

$\zeta(1 + \sec \alpha + M \tan \alpha)$ (see (b)): height at most h_T . At height Y its section has length $\omega + Y \hat{\kappa} \leq w_T$ (as $\hat{\kappa} \leq q_0^2/\zeta + \delta_Q$). Area at most A_T per complete block.

- Right end at s_0 , left end $x = 0$: for $j \leq \lfloor \chi \rfloor - 1$ the piece is placed ($j + 1 \leq \chi = r(0)$), waste at most $1 + \frac{1}{2} \tan \alpha_0$, in at most $\lfloor \chi \rfloor$ rows. The part of E_0 with $Y \geq \lfloor \chi \rfloor$ has $X < s_0 - \lfloor \chi \rfloor \tan \alpha_0 \leq (M_0 - \lfloor \chi \rfloor) \tan \alpha_0 < w_0$ and $Y \leq r(w_0) \leq \lfloor \chi \rfloor + 1 + \zeta w_0$: area at most $w_0(1 + \zeta w_0)$.
- Right end at $x = L$ (*final stretches*), at most one per row $j < \lfloor r(L) \rfloor$. If the left end is $x = 0$ or a slab line with $p \leq P - 1$, the piece is placed when $\ell \geq 1$ and the waste is at most $1 + \frac{1}{2} \tan \alpha_0$. Otherwise, by (c), the left end is on $\hat{\xi}$ of the last block, and these end rows lie in $E_{\text{end}} := \{\lambda_{\text{last}} \geq \hat{\xi}_{\text{last}}\}$, where $L - \hat{\xi}_{\text{last}} < 1 + \sec \alpha_0$ (the last block stopped with reason ‘ L ’, or the next block could not start). An end row with $a_j = \hat{\xi} - j \tan \alpha \geq 0$ and $j + 1 \leq r(a_j)$ is placed with waste at most $1 + \frac{1}{2} \tan \alpha_0$ (counted in $r(L)(1 + \frac{1}{2} \tan \alpha_0)$). The other points of E_{end} : (i) rows with $a_j \geq 0$ and $j + 1 > r(a_j)$; as $a_j \geq L - w_e$, these rows have $j > r(L) - \zeta w_e - 1$, and their points have $X \in [L - w_e, L]$, $Y \leq r(L)$: area at most $w_e(1 + \zeta w_e)$; (ii) rows with $a_j < 0$: then $j > \hat{\xi}/\tan \alpha \geq (s + \sec \alpha)/\tan \alpha \geq M + 1/\tan \alpha > N_0$ by (I2), so there are at most $r(L) - \lfloor \chi \rfloor + 1 \leq \zeta L + 2$ such rows, and $\hat{\xi} < j \tan \alpha \leq r(L) \tan \alpha_0$ gives $L < w_e$, so each such row meets Z in area at most $L < w_e$: area at most $(\zeta w_e + 2)w_e$. In total at most $w_e(4 + 2\zeta w_e)$.

(f4) Summing (the cells; per complete block, of which there are at most $\bar{K}$, the aligned, unaligned and transition rows; the start; the final stretches) gives $U_Z \leq B_W$. If there is no block at all ($s_0 + \sec \alpha_0 > L$), all rows run from 0 to L , rows $j \leq \lfloor \chi \rfloor - 1$ are placed with waste at most 1, and the rest has area at most $L(1 + \zeta L) \leq w_e(1 + \zeta w_e)$, as $L < s_0 + \sec \alpha_0 \leq w_e$. $\square$

8.2. The end packing. We use the band end $\text{Reg}(y)$ of Section 6 with $\tan(\theta/2) = b/(b^2 - 1)$, and, as in Corollary 7.4, $\mu := \tan \theta$, $g(X) := y + (X - \sigma)\mu$, $g_0 := g(0)$, $\bar{g} := g(w)$ and $\Delta := w\mu$.

Construction E38(y).

- (E1) $m := \lfloor \bar{g} \rfloor + 1$, $D_\ell := D_r := \lfloor 4m^{3/4} \rfloor$ and $g_D := g(D_\ell)$.
- (E2) *Main tilt:* $t := \lceil t^* Q \rceil / Q$ with $t^* = t^*(m, g_D)$ of Lemma 7.2 (g_0 replaced by g_D).
- (E3) *Main columns:* $\xi_i := D_\ell + m \tan \alpha + i \sec \alpha$ for $0 \leq i < N := \#\{i \geq 0 : \xi_i + \sec \alpha \leq w - D_r\}$, $\Gamma_i := g(\xi_i - m \tan \alpha) = g(D_\ell + i \sec \alpha)$, $J_i := \lfloor (\Gamma_i - m \cos \alpha - \sin \alpha)/q \rfloor$ and the columns $K(\xi_i, J_i, m - J_i)$. Put $J_L := J_{N-1}$ and $\xi_N := \xi_0 + N \sec \alpha$.
- (E4) *Cut:* $\kappa_m := q^2/\mu$; $\tau := \tan \alpha_c$, where t_c is the largest point of $(1/Q^*)\mathbb{Z}$ with $\tan \alpha_c \leq \tan \alpha - \kappa_m$; $\hat{\kappa} := \tan \alpha - \tau$. The phase ω_m is defined as in (Z3), with $(s_\beta, \hat{\xi}_\beta, \tan \alpha_{\beta+1}, B, A)$ replaced by $(\xi_0, \xi_{N-1}, \tau, q/(\mu \sec \alpha), (g_D - m \cos \alpha - \sin \alpha)/q)$ and the rows $J_0 \leq j < J_L$; put $\ell_0 := \xi_{N-1} + \omega_m$.
- (E5) *Main rows:* for $0 \leq j < J_L$ let $p(j) := \#\{i : J_i \leq j\}$, $a_j := \xi_{p(j)} - j \tan \alpha$ and $a_j^+ := \ell_0 - (j + 1)\tau$; if $a_j^+ - a_j \geq 1$ place $[a_j, a_j + \lfloor a_j^+ - a_j \rfloor] \times [j, j + 1]$.
- (E6) *Walls.* Each of the three regions below is filled by ZC with $t_0 := \lceil Q^* \tilde{t} \rceil / Q^*$, where $\tilde{t}$ is within relative error 10^{-12} of the tilt t_{tgt} with $\tan \alpha(t_{\text{tgt}}) = \frac{3}{2}\sqrt{\zeta}$ (since $d \log \tan \alpha / d \log t = (1 + t^2)/(1 - t^2)$, a relative error of at most 10^{-12} in t gives a relative error of at most $\epsilon' := 1.00001 \cdot 10^{-12}$ in $\tan \alpha$ whenever $t \leq 0.0014$, because then $(1 + t^2)/(1 - t^2) \leq 1.000004$), and mapped by an exact rotation by $\pm 90^\circ$ and a translation:
 - Z_L : $\chi = \xi_0 - g_0 \tan \alpha$, $\zeta = \tan \alpha$, $L = g_0$, $(x, y) \mapsto (y, g_0 - x)$; region $\{\lambda \leq \xi_0, Y \leq g_0\}$;
 - Z_{LR} : $\chi = w - \ell_0$, $\zeta = \tau$, $L = J_L$, $(x, y) \mapsto (w - y, x)$; region $\{X + Y\tau \geq \ell_0, Y \leq J_L\}$;
 - Z_{UR} : $\chi = w - \xi_N + J_L \tan \alpha$, $\zeta = \tan \alpha$, $L = Y_u - J_L$, $(x, y) \mapsto (w - y, J_L + x)$, with $Y_u := (g_0 + \mu \xi_N)/(1 + \mu \tan \alpha)$; region $\{\lambda \geq \xi_N, J_L \leq Y \leq Y_u\}$.

Here $\lambda = \lambda(X, Y) = X + Y \tan \alpha$ is the slab coordinate of the main tilt. Hypotheses: (E-i) $1 < g_D < m$ and $t \leq 1/10$; (E-ii) $N \geq 1$; (E-iii) $\tan \alpha - \kappa_m \geq \tan \alpha(1/Q^*)$, so that $\tau > 0$; (E-iv)

$\varepsilon_m + \tau < 1$, where $\varepsilon_m := (\sec \alpha - 1) + m\delta_Q + 1/Q_f$; (E-v) each of the three walls satisfies (W1)–(W5), for Z_{LR} and Z_{UR} with L replaced by the upper bound m (which only strengthens (W2) and (W3)).

Lemma 8.2 (end lemma for the exponent $3/8$). *Under (E-i)–(E-v), $E38(y)$ is a packing in $\text{Reg}(y)$, and its uncovered area satisfies*

$$U(y) \leq w \sin \alpha + wq + \Delta(\sec \alpha + m \tan \alpha) + J_0(1 + \tan \alpha) + J_L(\varepsilon_m + \tan \alpha) + \frac{1}{2}\mu h_L^2 + \mu w_R^2 + U_{Z_L} + U_{Z_{LR}} + U_{Z_{UR}},$$

where $h_L := \xi_0 - g_0 \tan \alpha$, $w_R := w - \xi_N + \bar{g} \tan \alpha$, and each $U_Z \leq B_W$ by Lemma 8.1.

Proof. (a) *Main staircase.* As in Lemma 7.3(a), with g_0 replaced by g_D and the shift D_ℓ : $\Gamma_i = g_D + i\mu \sec \alpha$ is affine in i ; $J_i \in [0, m-1]$, because $H(0) = m \cos \alpha + \sin \alpha \leq g_D \leq \Gamma_i$ by Lemma 7.2 and $H(m) = m + \sin \alpha > \bar{g} \geq \Gamma_i$; the J_i are non-decreasing; the gaps lie in $[0, q]$; and the columns lie in $\text{Reg}(y)$ ($X \geq \xi_i - m \tan \alpha = D_\ell + i \sec \alpha \geq 0$, $X \leq \xi_i + \sec \alpha \leq w - D_r$). The derivation of (R) in the proof of Lemma 8.1(b) applies verbatim to the rows $J_0 \leq j < J_L$ (with $p(j) = \lceil B(j+1-A) \rceil \in [1, N-1]$, $B(\sec \alpha - 1) = q^2/\mu = \kappa_m$, $\eta := \hat{\kappa} - \kappa_m \in [0, \delta_Q)$, $J_L - J_0 \leq m-1$, and $\hat{\xi}$ replaced by ξ_{N-1}): the length $\ell_j := a_j^+ - a_j$ satisfies $\text{frac}(\ell_j) \leq \varepsilon_m$. Moreover $\ell_j \geq \ell_0 - \xi_{p(j)} - \tau \geq -\tau$ (as $p(j) \leq N-1$); together with $\text{frac}(\ell_j) \leq \varepsilon_m < 1 - \tau$ (E-iv) this forces $\ell_j \geq 0$. The rows $j < J_0$ ($p(j) = 0$) are not aligned and waste at most $1 + \tan \alpha$ each. Row ceiling: for $j < J_L$, $p(j) \leq N-1$ and $J_{p(j)} \geq j+1$, so $j+1 \leq H(J_{p(j)}) \leq \Gamma_{p(j)} = g(\xi_{p(j)} - m \tan \alpha) \leq g(a_j)$ (as $j < m$). Also $a_j \geq \xi_0 - m \tan \alpha = D_\ell \geq 0$ and $a_j^+ \leq \ell_0 \leq w$.

(b) *The wall regions lie in $\text{Reg}(y)$.* $Z_L = \{0 \leq Y \leq g_0, 0 \leq X \leq \xi_0 - Y \tan \alpha\}$ and $Y \leq g_0 \leq g(X)$. $Z_{LR} = \{0 \leq Y \leq J_L, \ell_0 - Y\tau \leq X \leq w\}$: its points have $X \geq \ell_0 - J_L\tau \geq \xi_{N-1} - m \tan \alpha$, so $g(X) \geq \Gamma_{N-1} \geq H(J_L) \geq J_L \geq Y$. $Z_{UR} = \{J_L \leq Y \leq Y_u, \xi_N - Y \tan \alpha \leq X \leq w\}$: by the definition of Y_u , $Y_u = g(\xi_N - Y_u \tan \alpha)$, so $Y \leq Y_u \leq g(\xi_N - Y \tan \alpha) \leq g(X)$; and $J_L \leq \Gamma_{N-1} \leq g(\xi_N - J_L \tan \alpha)$ gives $J_L \leq Y_u$ (the map $Y \mapsto g(\xi_N - Y \tan \alpha) - Y$ is decreasing), that is $L \geq 0$. The maps are isometries, so by Lemma 8.1 every wall piece lies in its region.

(c) *Disjointness.* Main columns: slabs. Main rows against columns: as in Lemma 7.3(e) on the left (the line $\lambda = \xi_{p(j)}$, and columns with $J \geq j+1$ above $Y = j+1$); on the right the rows lie in $\{X + Y\tau \leq \ell_0\}$, the columns $i \leq N-2$ in $\{\lambda \leq \xi_{N-1}\} \subset \{X + Y\tau \leq \ell_0\}$ (as $\tau \leq \tan \alpha$, $Y \geq 0$ and $\ell_0 \geq \xi_{N-1}$), and column $N-1$ lies in $\{Y \geq J_L\}$ while the rows lie in $\{Y \leq J_L\}$. $Z_L \subset \{\lambda \leq \xi_0\}$, and all main pieces lie in $\{\lambda \geq \xi_0\}$. $Z_{LR} \subset \{X + Y\tau \geq \ell_0\}$, against the main rows and the columns $i \leq N-2$; against column $N-1$, Z_{UR} and the main rows the line $Y = J_L$ is used where needed. $Z_{UR} \subset \{\lambda \geq \xi_N\}$, against the main columns (in $\{\lambda \leq \xi_N\}$), and against the rows by $Y = J_L$. Z_L against Z_{LR} : Z_{LR} lies in $\{\lambda \geq \xi_{N-1} \geq \xi_0\}$. Z_L against Z_{UR} : $\lambda = \xi_0 \leq \xi_N$. Z_{LR} against Z_{UR} : $Y = J_L$.

(d) *Accounting.* Partition $\text{Reg}(y)$ into $\{\lambda < \xi_0\}$ ($= Z_L$ plus $S_1 := \{\lambda < \xi_0, Y > g_0\}$); $\{\lambda \geq \xi_N, Y \geq J_L\}$ ($= Z_{UR}$ plus S_2 , its part with $Y > Y_u$); $\{X + Y\tau > \ell_0, Y < J_L\}$ ($= Z_{LR}$ up to the boundary); the column cells $\{\xi_i \leq \lambda < \xi_{i+1}, Y \geq J_i\}$ minus the previous set (for $i \leq N-2$ the cell does not meet it; for $i = N-1$, $Y \geq J_L$); and the row regions $\{j \leq Y < j+1, \lambda \geq \xi_{p(j)}, X + Y\tau \leq \ell_0\}$, $0 \leq j < J_L$. The cells cost at most $w \sin \alpha + wq + \Delta(\sec \alpha + m \tan \alpha)$, exactly as in Lemma 7.3(g) ($N \sec \alpha \leq w$, $N \tan \alpha \leq w \sin \alpha$). A row region has sections of length $\ell_0 - \xi_{p(j)} + Y\hat{\kappa}$, so its area is at most $\ell_j + \frac{1}{2}(\tan \alpha + \tau)$ and its waste at most $\varepsilon_m + \tan \alpha$ for $J_0 \leq j < J_L$, and at most $1 + \tan \alpha$ for $j < J_0$. S_1 consists of points with $0 \leq X < \xi_0 - g_0 \tan \alpha = h_L$ and $g_0 < Y \leq g_0 + \mu X$: area at most $\frac{1}{2}\mu h_L^2$. S_2 consists of points with $X \in [\xi_N - \bar{g} \tan \alpha, w]$ and $Y_u < Y \leq g(X) \leq Y_u + \mu w_R$: area at most μw_R^2 . The three wall regions contribute $U_{Z_L} + U_{Z_{LR}} + U_{Z_{UR}}$ (the maps are isometries). $\square$

8.3. The band ends.

Corollary 8.3. *Let $b \geq b_0 := 10^{16}$ be an integer, $\tan(\theta/2) = b/(b^2 - 1)$, and let y be any real number with*

$$\frac{1}{12}b^{4/5} \leq y \leq \frac{1}{12}b^{4/5} + 2.$$

Let Q be a power of 2 with $Q \geq 16b$. Then $E38(y)$ is a packing in $\text{Reg}(y)$ with

$$U(y) \leq C_E b^{3/5}, \quad C_E := 15.54,$$

and the end cost of Proposition 6.3 satisfies $E(y) = U(y) + \frac{1}{2} \tan \theta \leq 15.54 b^{3/5} + 1.0001/b$.

Proof. Put $z := m^{-1/4}$.

(i) *Ranges.* By the band facts of Corollary 7.4(i), $2/b < \mu \leq 2.000001/b$, $\Delta \leq 2.000001$, $\sigma\mu \leq 4.1 \cdot 10^{-8}$ and $b - 2.1/b \leq w < b$. As $\bar{g} = y - \sigma\mu + \Delta \in (y, y + 2.000001]$ and $m = \lfloor \bar{g} \rfloor + 1$, we get $\frac{1}{12}b^{4/5} < m \leq \frac{1}{12}b^{4/5} + 5.000001$, $\delta := m - g_0 \leq 3.000001$ and $m - g_D \leq \delta$. So $\hat{b} := bz^5$ lies in $[12^{5/4}(1 - 5.000001z^4)^{5/4}, 12^{5/4}]$ and $z < 12^{1/4}b^{-1/5} \leq z_0 := 12^{1/4}10^{-16/5} = 0.0011743\dots$; also $D_\ell z^3, D_r z^3 \in [4 - z^3, 4]$. For the main tilt, Lemma 7.2 with $D' = m + g_D \geq 2m - 3.000001$, $\delta' = m - g_D \leq 3.000001$, and $t^* \geq \sqrt{\delta'/D'}$ with $D' \leq 2m$ and $\delta' > \bar{g} - g_D = (w - D_\ell)\mu \geq 2(1 - D_\ell/b - 2.1/b^2)$, give

$$\frac{t}{z^2} \in \left[\sqrt{1 - D_\ell/b - 2.1/b^2}, \frac{z^2 + \sqrt{z^4 + 3.000001(2 - 3.000001z^4)}}{2 - 3.000001z^4} + \frac{z^3}{16\hat{b}} \right].$$

Further $\kappa_m = q^2/\mu \leq q^2b/2$ and $\tau \in [\tan \alpha - \kappa_m - \delta_Q, \tan \alpha - \kappa_m]$. Wall inputs: Z_L : $\zeta = \tan \alpha$, $\chi \in [D_\ell, D_\ell + 3.000001 \tan \alpha]$, $L \leq m$; Z_{LR} : $\zeta = \tau$, $\chi \in [D_r + \sec \alpha - 1, D_r + 2 \sec \alpha]$, $L = J_L$; Z_{UR} : $\zeta = \tan \alpha$, $\chi \in [D_r, D_r + \sec \alpha + m \tan \alpha]$, $L = Y_u - J_L$; and $J_L + (Y_u - J_L) = Y_u \leq \bar{g} < m$. Wall tilts: the target tilts satisfy $\tan \alpha(t_{\text{tgt}}) = \frac{3}{2}\sqrt{\zeta} \leq \frac{3}{2}\sqrt{\tan \alpha} \leq 0.00276$ (as $\zeta \leq \tan \alpha \leq 3.38 \cdot 10^{-6}$, see (iii)), so $t_{\text{tgt}} \leq 0.0014$, and by (E6) $\tan \alpha_0 \in [\frac{3}{2}\sqrt{\zeta}(1 - \epsilon'), \frac{3}{2}\sqrt{\zeta}(1 + \epsilon') + \delta_Q]$ with $\epsilon' = 1.00001 \cdot 10^{-12}$. (E-i) is immediate: $m - 3.000001 \leq g_0 \leq g_D \leq \bar{g} < m$.

(ii) *Normalisation.* Every term of Lemma 8.2 and of B_W (with $\varpi = 0.65$) is multiplied by $z^3 = m^{-3/4}$ and rewritten in terms of $\hat{b}$, t/z^2 , $\tan \alpha/z^2$, ζ/z^2 , χz^3 , $Lz^4 \in [0, 1]$, $\tan \alpha_0/z$, q_0/z^2 , $q_{\min}/z^2$, $N_0 z^3$ and Λz^3 , so that only non-negative powers of z occur; the complete list of normalised enclosures is in Appendix A and in the program `cert3e.py` (for example $w \sin \alpha z^3 \leq 2\hat{b}(t/z^2)$, $\bar{K} = (Lz^4)/(\Lambda z^3) \cdot z^{-1}$, and $J_0 \leq (4mt + 2)/(Qq)$, because $H(0) = m \cos \alpha + \sin \alpha$ has $|dH/dt| \leq 4mt + 2$ and $t - t^* \leq 1/Q$). For the two right walls the terms linear in L are bounded with the larger of the two coefficients (as $L_{LR} + L_{UR} \leq m$) and the other terms are evaluated at $L = m$ (B_W and the hypotheses (W2), (W3) are monotone in L).

(iii) *Interval evaluation.* $[0, z_0]$ is split into intervals and the range of t/z^2 into pieces; on each box all enclosures are evaluated in interval arithmetic. With 160×64 boxes and 120-bit intervals the lower bounds of the margins are: $1/10 - t \geq 0.0999$; (E-ii) ≥ 22.33 (normalised by z^5); (E-iii) ≥ 1.946 (normalised by z^2); $1 - \varepsilon_m \geq 0.99999$; for each wall (W1) ≥ 1.88 , (W2) ≥ 0.1314 (normalised by z), (W3) ≥ 3.998 , (W4) ≥ 0.99999 , (W5) ≥ 1.36 (normalised by z) and $1/10 - t_0 \geq 0.097$; and $\Psi(z) := z^3 \times (\text{the bound of Lemma 8.2})$ satisfies $\Psi \leq 96.477$. Directly, $\varepsilon_m \leq 6.91 \cdot 10^{-6}$ (as $\sec \alpha - 1 \leq 5.8 \cdot 10^{-12}$, $m\delta_Q \leq \frac{2.1}{16} \cdot \frac{m}{b} \leq 6.902 \cdot 10^{-6}$ and $1/Q_f \leq 2^{-48}$) and $\tau \leq \tan \alpha \leq 3.38 \cdot 10^{-6}$ (as $t \leq 1.2248z_0^2$), so $\varepsilon_m + \tau \leq 1.03 \cdot 10^{-5} < 1$: (E-iv) holds.

(iv) By Lemmas 8.2 and 8.1, $U(y) \leq \Psi_{\text{sup}} m^{3/4} \leq \Psi_{\text{sup}} (\frac{1}{12}b^{4/5} + 5.000001)^{3/4} \leq \Psi_{\text{sup}} 12^{-3/4} (1 + 60.000012b_0^{-4/5})^{3/4} b^{3/5}$, and with $\Psi_{\text{sup}} \leq 96.477$ this is at most $14.964 b^{3/5} \leq 15.54 b^{3/5}$. The relation between E and U is as in Corollary 7.4(vi). $\square$

Certification. The author's own program (`mpmath.iv`, 40×16 boxes) gave $\Psi \leq 100.1875$ and $C_E \leq 15.5392$; this is the origin of $C_E = 15.54$. It used the relative slack 10^{-12} on the wall tilts, which is too little ((E6) requires $\epsilon' = 1.00001 \cdot 10^{-12}$), so it does not count as a certification. The constants in (iii) and (iv), and $C_E = 15.54$, are certified by three independent full-range coverings written from the text of the proof, with different libraries: an independent reimplementaion (`mpmath.iv`, 120 bits, 160×64 boxes, slack $1.03 \cdot 10^{-12}$; $C_E \leq 14.964$, the values quoted in (iii); this is the program `cert3e.py` included with this text), a review (`mpmath.iv`, 200×64 boxes, slack $1.00001 \cdot 10^{-12}$; $C_E \leq 14.963$), and a second review (`python-flint arb`, slack $1.0001 \cdot 10^{-12}$; $C_E \leq 14.9631$). All

three use a slack $\epsilon' \geq 1.00001 \cdot 10^{-12}$ and give $C_E \leq 14.964$. In the two `mpmath.iv` coverings the smallest normalised margin is that of (W2). At the worst box of the 160×64 covering the bound splits as column triangles 54.71, other main terms 0.01, left wall 16.61 and right walls 25.14 (in units of $m^{3/4}$). Exponent bookkeeping: the triangles cost about $bt \asymp bm^{-1/2} \asymp m^{3/4}$ (as $m \asymp b^{4/5}$), and each wall about $L \tan \alpha_0 + \bar{K} N_0 \tan \alpha_0 + \bar{K}/q_{\min} + 2\chi \asymp m^{3/4}$, with $\tan \alpha_0 \asymp m^{-1/4}$, $\chi \approx 4m^{3/4}$ and $\bar{K} \asymp m^{1/4}$.

Why $b_0 = 10^{16}$. Hypothesis (E-iii) ($\tau > 0$, where $\kappa_m/\tan \alpha \approx 76b^{-1/5}$) fails below about $b = 3 \cdot 10^9$, and for smaller b_0 the certified constant grows (about 16.7 for $b_0 = 10^{12}$ and 19.2 for $b_0 = 10^{10}$). As $b \rightarrow \infty$ the certified bound of this proof tends to about 14.6 (a covering with a fine subdivision of the t -range and $b_0 = 10^{60}$ gives 14.61); this reflects the crude estimates of Lemma 8.1, not $c^*(k)$.

8.4. Proof of Theorem 1.1. Let $C_E := 15.54$, $k \geq k_1 := 4 \cdot 10^{26}$, $b_* := (5k/(3C_E))^{5/8}$ (so $k = \frac{3}{5}C_E b_*^{8/5}$) and $b := \lceil b_* \rceil$. Since $\frac{3}{5}C_E \cdot 10^{128/5} \leq 3.712 \cdot 10^{26} \leq k_1$, $b \geq b_* \geq 10^{16} = b_0$. Take $\tan(\theta/2) = b/(b^2 - 1)$, so $w < b$ and $S := k - b + w < k$, and use the packing of Corollary 6.2 with

$$y_0 := \frac{1}{4} \lceil \frac{1}{3} b^{4/5} \rceil \in [\frac{1}{12} b^{4/5}, \frac{1}{12} b^{4/5} + \frac{1}{4}]$$

and, in each band, $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$. As $h < 3$ and $y_0 \leq k^{1/2} + 1$, $h + 2y_0 < k - b \leq L_b$; so $R \geq 1$ and $y_1 \in [y_0, y_0 + d)$ with $d < 1.0001$, and both lifts lie in the window of Corollary 8.3. Fill the four end regions by Corollary 8.3 with Q the least power of 2 that is at least $16b$. As in Section 7.3, by Corollary 6.2 and Proposition 6.3,

$$k^2 - N < \frac{4k}{b} + 4C_E b^{3/5} + \lambda', \quad \lambda' := \frac{4k}{b(b^2 - 1)} + \frac{4kb}{(b^2 - 1)^2} + 2 \tan \theta \leq \frac{8.001 k}{b^3} + \frac{4.1}{b}.$$

With $k \leq \frac{3}{5}C_E b^{8/5}$ we get $8.001 k/b^3 \leq 8.001 \cdot 9.324 b^{-7/5} < 10^{-20}$ and $4.1/b < 10^{-15}$, so $\lambda' < 1$, and Lemma 2.2 gives $c^*(k) \leq k^2 - N - 1 < 4k/b + 4C_E b^{3/5}$. Finally $4k/b \leq 4k/b_* = \frac{12}{5}C_E b_*^{3/5}$ and $b^{3/5} \leq b_*^{3/5} + \frac{3}{5}b_*^{-2/5} \leq b_*^{3/5} + \frac{3}{5} \cdot 10^{-32/5}$, so

$$c^*(k) < \frac{32}{5}C_E b_*^{3/5} + \frac{12}{5}C_E \cdot 10^{-32/5} = \frac{32}{5} \left(\frac{5}{3}\right)^{3/8} C_E^{5/8} k^{3/8} + \frac{12}{5}C_E \cdot 10^{-32/5}.$$

In interval arithmetic $\frac{32}{5}(\frac{5}{3})^{3/8} 15.54^{5/8} \in [43.05564, 43.05565]$ and $\frac{12}{5} \cdot 15.54 \cdot 10^{-32/5} \leq 1.49 \cdot 10^{-5}$, which gives Theorem 1.1. $\square$

Remark 8.4 (what the explicit computations test). The asymptotic regime of Corollary 8.3 ($b \geq 10^{16}$, $k \geq 4 \cdot 10^{26}$) cannot be tested by explicit computation: one end has about 10^{16} pieces. The constants are certified by interval arithmetic only. The explicit checks test the validity of the construction and of Lemmas 8.1 and 8.2 at smaller b . The construction was implemented in exact rational arithmetic and checked by an exact checker that does not use the lemmas (shapes, containment, separating axis tests for all pairs with overlapping bounding boxes): for $b = 1600, \dots, 409600$ and five lifts each in the window of Corollary 8.3 (all 25 pass; there the cut is not aligned, as (E-iii) needs $b \gtrsim 3 \cdot 10^9$, so these runs test validity only), at lifts outside the window where the cut is aligned (all pass, with the residuals of (R) attained to 4–5 digits), and for the wall filler alone on $Z(4m^{3/4}, \zeta, m)$ for m up to 10^4 (certified) and $2.56 \cdot 10^6$ (exact internal checks of Lemma 8.1), with $U_Z \leq B_W$ in every case. Deliberately corrupted packings were rejected. Two blank-slate AI reviews and an independent reimplement, each with its own programs written from the text, found no counterexample (over 500 wall fillers, chains of up to 118 blocks, end packings up to $b = 10^7$). The program `run_z.py` included with this text builds the wall filler for $m = 10^4$, $\zeta = 0.5/\sqrt{m}$, $\chi = 4m^{3/4}$, $L = m$ and certifies it: 3 blocks, 25289 pieces, 28402 pairs decided exactly, no overlap, (W1)–(W5) hold, 7611 aligned closings with largest residual $0.0056087 \leq \varepsilon_W = 0.0056093$, and $U_Z = 6712 \leq B_W \leq 19615$. Full exact counts with the construction of this section give $c^*(10^8) \leq 44299$, $c^*(10^{10}) \leq 236005$ and

$c^*(10^{11}) \leq 547881$ (ends checked exactly); at $k = 10^8$ this is worse than Theorem 1.6 (24790), and at such k the closed form $43.06 k^{3/8}$ must not be used.

9. THE CERTIFIED PACKINGS

Generation. The end packings of Theorem 1.6 were produced by a variant of the construction of Section 7.1 with tuned parameters: b was chosen near the minimizer of a model of the total cost, and the lift y_0 (about $0.9b^{2/3}$) by a floating-point pre-scan; for each end the column length m is $\lfloor \bar{g} \rfloor$ or $\lfloor \bar{g} \rfloor + 1$, whichever leaves less uncovered area; the tilt is the least multiple t of 2^{-40} with $m \cos \alpha + \sin \alpha \leq g(\cos \alpha)$ (the top vertex of the first column, which starts at the left wall, has abscissa $\cos \alpha$); and each J_i is the largest value, at least J_{i-1} , for which the column (of $m - J_i$ squares) fits under the ceiling. These rules play no role in the verification, which uses only the resulting explicit pieces.

Proof of Theorem 1.6. Let (k, b, y_0) be a row of Table 1 and $\tan(\theta/2) = b/(b^2 - 1)$. Then $\tan(\theta/2) > 1/b$, so $\theta \in (\theta_0, \pi/2)$ and $w < b$ (Lemma 3.2), and γ, σ are rational. For both bands let $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$, so that $y_0 \leq y_1 < y_0 + d$. The data file for k lists, for each of the four ends, its lift (y_0 for the lower ends, y_1 for the upper ends) and its pieces, with exact rational coordinates in the frame of $\text{Reg}(y)$: *columns* (a tilted $1 \times n$ rectangle, given by its lowest vertex, n and the rational cosine and sine of its tilt), *layers* and *wedge rows* (axis-parallel $1 \times L$ rectangles). By Corollary 6.2 and Lemma 2.2 it suffices to check, in exact rational arithmetic:

- (C0) the lift given in the file for each end equals y_0 (lower ends) or the computed y_1 (upper ends), and every n and L is a positive integer;
- (C1) $w < b$, $R \geq 1$, $y_0 \geq 0$ and $y_1 \geq 0$ for both bands;
- (C2) every piece is a rectangle with side lengths 1 and n (hence a union of n unit squares with pairwise disjoint interiors); for a column this means $\cos^2 + \sin^2 = 1$ exactly;
- (C3) the four vertices of every piece lie in $\text{Reg}(y)$ (which is convex);
- (C4) two pieces of the same end have disjoint interiors.

For (C4) it suffices to test pairs whose bounding boxes overlap, and for such pairs the separating axis test along the edge normals of the two rectangles is exact (touching is allowed). The count $N = (k - b)^2 + b(R_{\text{right}} + R_{\text{top}}) + (\text{end squares})$ gives the values $c = k^2 - N$ of Table 1, and Table 2 lists the resulting data. All checks pass (see below). $\square$

k	R_{right}	R_{top}	pieces (4 ends)	body cost	end cost
10^5	99868	99245	2887	639.20	944.79
10^6	999671	997190	10111	1609.72	2363.28
10^7	9999166	9989291	42550	4048.29	5991.71
$3.825 \cdot 10^7$	38248577	38226491	93680	6925.21	10063.79
10^8	99997915	99958601	165152	10172.28	14618.72

TABLE 2. Data of the certificates. Body cost is $\sum(R - 1) \tan \theta$ over the two bands and end cost is the sum of the four $E(y)$ of Proposition 6.3; their sum is c up to $k^2 - S^2 < 0.002$ (rounded to two decimals). The average of $E/b^{2/3}$ over the four ends is between 3.1 and 3.3.

Verification. Three programs check the certificates in exact rational arithmetic.

- (a) The generating program checks, for each end, that all vertices lie in $\text{Reg}(y)$, that each column lies in its slab $\{\xi_i \leq \lambda \leq \xi_i + \sec \alpha\}$ and above height J_i , that the slabs do not overlap, that the J_i are non-decreasing, that each layer starts to the right of the slabs of the columns below it and each wedge row to the left of the first slab (the separating lines of

Lemma 7.3(e)), and the exact uncovered area; it also checks the identity of Proposition 6.3 exactly.

- (b) A checker written from a description of the construction, without reading the generating program, takes only the parameters of the pieces from it, builds the vertices itself and checks (C0)–(C4) with the separating axis test for all pairs of pieces with overlapping bounding boxes, the exact uncovered area, and the count. It reproduced N and c for all five rows (for $k \geq 10^7$ on a remote machine), and rejected deliberately corrupted end packings for $b = 400$ and 6400 (a column shifted, lowered or lengthened, a layer raised, an extra wedge square, a lowered ceiling).
- (c) The checker `stair_check.py`, included with this text, reads only the published data files. It recomputes the band, the rows R and the upper lifts y_1 from (k, b, y_0) , checks (C0)–(C4) with the separating axis test for all pairs of pieces with overlapping bounding boxes, the exact uncovered areas, the count and the identity of Proposition 6.3. It was adapted from checker (b) and does not import the generating program. Its results are in Table 3; seven deliberately corrupted versions of the data (a column shifted, lowered or given an extra square, a layer raised or lengthened, a wedge row lengthened, the ceiling lowered by $1/100$) were all rejected at $k = 10^5$ and at $k = 10^8$.

k	pieces	pairs decided exactly	overlaps	result (time)
10^5	2887	22274	0	pass (0.3 s)
10^6	10111	119522	0	pass (1.2 s)
10^7	42550	969664	0	pass (8.2 s)
$3.825 \cdot 10^7$	93680	2730501	0	pass (29.6 s)
10^8	165152	5748521	0	pass (59.2 s)

TABLE 3. Results of the checker `stair_check.py` (c), summed over the four ends, with Python `fractions` on a laptop. For $k \geq 10^7$ it was also run with `gmpy2` rationals on a remote Linux machine, with the same output apart from the arithmetic and timing lines.

All checkers were written by AI systems; none has been reviewed by a human. The rows of the bands are not listed in the data; their correctness rests on Lemmas 3.3 and 6.1.

The tuned packings are 4.9–11.8% better than the packings of the proof of Theorem 1.2 (Remark 7.5); for $k = 10^5, \dots, 10^8$ the ratio $c/k^{2/5}$ is between 15.6 and 16.0, and the local slopes of $\log c$ against $\log k$ between 10^5 , 10^6 , 10^7 and 10^8 are 0.399, 0.403 and 0.393.

10. COMPARISON WITH THE LITERATURE

10.1. From the wasted area to the deficiency. By Lemma 2.1, an upper bound $W(x) \leq F(x)$ that holds for all x in an interval $(k-\delta, k)$ gives $c^*(k) \leq \limsup_{x \rightarrow k^-} F(x) - 1$, that is $c^*(k) \leq F(k) - 1$ if F is continuous. A bound that holds only for some x (for instance for x whose fractional part is bounded away from 0) gives nothing for $c^*(k)$. Conversely, a packing that shows $c^*(k) \leq c - 1$ shows $W(x) \leq x^2 - k^2 + c$ for x slightly below k .

10.2. Printed bounds. The bounds in the literature that we found are listed below (literature search of 2026-10-08; the last column is what they give for $c^*(k)$ by the translation above).

source	statement	consequence for $c^*(k)$
Erdős–Graham 1975 [EG]	$W(x) = O(x^{7/11})$, all real x	$O(k^{7/11})$
Chung–Graham 2009 [CG]	$W(x) \leq Cx^{(3+\sqrt{2})/7} \log x$	$O(k^{(3+\sqrt{2})/7} \log k)$, exponent ≈ 0.6306
Wang–Dong–Li 2016 [WDL] (arXiv)	$W(x) \leq (16\sqrt{2} + 38)x^{5/8}$	$\leq 60.63 k^{5/8} - 1$
Arslanov–Mustafin–Shangitbayev 2021 [AMS]	$s(n^2 - n) < n$ for $n > 12$	$\leq k - 1$ for $k > 12$
Bui 2025 [Bui], McClenagan 2026 [McC] (preprints)	$W(x) = O(x^{3/5})$, all real x	$O(k^{3/5})$, implicit constant

The best printed bound for $c^*(k)$ is therefore $O(k^{3/5})$. The statement $W(x) = O(x^{3/5})$ was first claimed by Chung and Graham [CG2]; an error in that argument is pointed out in [Bui] and [McC]. We have checked the statement of [WDL, Theorem 1] but not whether every step of its proof applies to x just below an integer. The bound $c^*(k) = O(k^{3/5})$ is stated explicitly in the online problem collections [Ev] and [Le]. [AMS] is the only refereed source we found that states a bound in the form of $c^*(k)$; it also notes that [EG] gives $s(n^2 - O(n^{7/11})) < n$. Göbel [Gö] remarked that an improvement of the trivial bound is possible for large n in each range $k^2 + k \leq n < (k+1)^2$, without an order.

For $k = 10^5$ the explicit bound of [WDL] gives about $8.1 \cdot 10^4$, Theorem 1.4 gives 2535 and Theorem 1.6 gives 1583.

For small k , the Squares-in-Squares page maintained by D. Ellsworth [El] (web resource, version of 2026-09-24) lists packings showing $s(n^2 - n + 1) < n$ for $n = 16, 17, 18$, that is $c^*(k) \leq k - 2$ for these k .

To the best of our knowledge, no upper bound of order $o(k^{3/5})$ for $c^*(k)$, or for $W(x)$ just below an integer, has been stated in the searched sources; in particular we found neither $c^*(k) = O(\sqrt{k})$ nor any statement about the ends of tilted bands. The search covered [EG], [RV], [CG], [CG2], [WDL], [Bui] (versions 1 and 2), [McC] and the thesis [McC2], [AMS], Friedman’s survey [Fr] and the Squares-in-Squares page [El], the repositories [Le] and [Ev], MathOverflow and Mathematics Stack Exchange, and arXiv up to 2026-10-08. Neither [Bui] nor [McC] makes a special statement for sides just below an integer. A search cannot exclude unpublished or oral results. The ingredients of Theorem 1.3 are standard, and Theorem 1.4 may well be regarded as an easy consequence of known techniques; the staircase of Section 7 is the primitive construction of [Bui, Section 3] (Section 1.2).

10.3. Why uniform bounds cannot go below $\sqrt{k}$. Roth and Vaughan [RV, Theorem] proved that $W(x) \gg (x\|x\|)^{1/2}$ whenever $x(x - \lfloor x \rfloor) > 1/6$, where $\|x\|$ is the distance from x to the nearest integer.¹ In particular $W(n + \frac{1}{2}) \gg n^{1/2}$. Hence an upper bound $W(x) \leq Cx^\alpha$ valid for all large real x must have $\alpha \geq 1/2$, and through (1.1) such a bound can never give $c^*(k) = O(k^\alpha)$ with $\alpha < 1/2$. A bound of order $o(\sqrt{k})$ for $c^*(k)$ must use the fact that the side is just below an integer, where the theorem of Roth and Vaughan gives nothing ($\|x\| \rightarrow 0$). The constructions of this paper do so: the width $b - \varepsilon$ of the bands is just below the integer b . Erdős and Graham remark that perhaps the correct bound is $W(x) = O(x^{1/2})$ [EG, p. 122]; Theorem 1.4 gives the corresponding order for $c^*(k)$, and Theorems 1.2 and 1.1 go below it, just below integers.

11. LOWER BOUNDS AND THE ORDER OF $c^*(k)$

In the notation of Section 2, $c^*(k) = W_{\min}(k) - 1$. The following lower bounds are in preprints by the author that have not been refereed.

¹The theorem is sometimes quoted with $x - \lfloor x \rfloor$ in place of $\|x\|$, for example on p. 1 of [McC]. That form would give $W(x) \gg \sqrt{x}$ for x just below an integer, and hence $c^*(k) \gg \sqrt{k}$, contradicting Theorem 1.2. The original statement [RV, p. 170] has $\|x\|$ and gives nothing for x just below an integer.

- [R]: $W_{\min}(k) \geq 0.0353 \log k$ for every integer $k \geq 2$, where $\log$ is the natural logarithm (version 1.2; the constant was 0.033 in earlier versions). Thus $c^*(k) \rightarrow \infty$. The constant uses a computer-assisted lemma ([R, Lemma 4.10]).
- [Q]: $W_{\min}(k) \geq 0.1 k^{1/4}$ for every integer $k \geq 4.62 \cdot 10^{12}$.
- [T]: $W_{\min}(k) > 0.06 k^{1/3}$ for every integer $k \geq 3.5 \cdot 10^{17}$, and $\liminf_{k \rightarrow \infty} W_{\min}(k)/k^{1/3} \geq 0.06285$. Without [R, Lemma 4.10], $W_{\min}(k) > 0.05 k^{1/3}$ for $k \geq 3.39 \cdot 10^{17}$.

For small k see [R, Section 1]: $W_{\min}(k) \geq 3$ for $k \geq 2$, that is $c^*(k) \geq 2$, and a computer-assisted result of E. Daniel that has not been refereed gives $s(k^2 - 3) = k$ for $k \geq 6$, that is $c^*(k) \geq 3$ ([R, Corollary 1.3]; [Le, entry T-064]).

Together with Theorems 1.2 and 1.1,

$$\begin{aligned} 0.06 k^{1/3} - 1 &< c^*(k) < 20.668 k^{2/5} + 0.623 & (k \geq 3.5 \cdot 10^{17}), \\ c^*(k) &< 43.06 k^{3/8} + 2 \cdot 10^{-5} & (k \geq 4 \cdot 10^{26}), \end{aligned}$$

so $k^{1/3} \ll c^*(k) \ll k^{3/8}$ is what is proved: the upper bound by Theorem 1.1 of this paper and the lower bound in [T]. Neither exponent is claimed to be optimal; the true exponent is open (Section 12). The gap is large for moderate k : for $k = 10^8$ the lower bounds above give only $c^*(k) \geq 2$ ($c^*(k) \geq 3$ with the result of Daniel), while Theorem 1.6 gives $c^*(k) \leq 24790$.

12. OTHER END CONSTRUCTIONS AND OPEN PROBLEMS

Nothing in this section is used elsewhere. By Proposition 6.3, an end packing with uncovered area $O(b^\beta)$, uniformly for lifts in an interval of length $\sec \theta$, gives $c^*(k) \leq \min_b (4k/b + O(b^\beta)) = O(k^{\beta/(1+\beta)})$, with $b \asymp k^{1/(1+\beta)}$. The staircase has $\beta = 2/3$, and the chained wall fillers of Section 8 have $\beta = 3/5$.

Remark 12.1 (an intermediate construction: tilted columns). Before the staircase, we filled the end regions by columns of n squares whose tilt is chosen separately for each column from the fractional part of the ceiling height, so that $n \cos \beta + \sin \beta$ matches the ceiling; neighbouring columns are separated by slabs as in Lemma 7.1(i), but with an extra gap whenever the tilt increases. The loss under the ceiling is then about $\sqrt{2/M}$ per unit width for columns of height M , and each change of tilt costs a gap of order $M \tan \beta$; with $M \asymp \sqrt{b}$ this gives uncovered area $O(b^{3/4})$ and $c^*(k) = O(k^{3/7})$. The staircase avoids the changes of tilt altogether: one tilt serves all columns, and the rising ceiling is followed by moving squares into the layers, each move raising a column by $1 - \cos \alpha$. Since Theorem 1.2 has a smaller exponent, and its proof does not use this construction, we do not include the details or a proof of the $k^{3/7}$ bound.

Remark 12.2 (Type 3 regions of Chung and Graham). The end regions $\text{Reg}(y)$ are right trapezoids of width $w \approx b$ whose two parallel sides differ by about 2. Cut by two vertical lines into three trapezoids of width $w/3$, and with lifts y of order b , each piece appears to satisfy the hypotheses of the “Type 3” trapezoids of [CG, Theorem 1(3)], for which a packing with uncovered area $O(x^{1/\sqrt{2}} \log x)$ is asserted there. Combining Proposition 6.3 with [CG, Theorem 1(3)] therefore appears to give $c^*(k) = O(k^{\sqrt{2}-1} (\log k)^{2-\sqrt{2}})$. The exponent $\sqrt{2}-1 \approx 0.414$ is larger than $2/5$ and than $3/8 = 0.375$, so this route is weaker than Theorems 1.2 and 1.1. We do not verify the details, and the proof in [CG] is written in outline, with lower-order terms suppressed.

Remark 12.3 (the wall wedges). In the bound of Lemma 7.3, the column triangles cost about $w \sin \alpha \asymp b/\sqrt{m}$ and the two walls (the wedge rows and the ends of the layers) cost about m each, with $m \asymp b^{2/3}$. Section 8 reduces the cost of a wall of height m from order m to order $m^{3/4}$ (following ideas of [Bui, Sections 3–4]); then the best column length is $m \asymp b^{4/5}$, the end cost is $O(b^{3/5})$, and $c^*(k) = O(k^{3/8})$. The constants of Section 8 are crude (the parameters 4, $3/2$, 0.65 and $1/12$ and several per-row bounds were not optimised).

Open problems. The true exponent of $c^*(k)$ is open: it lies between $1/3$ ([T]) and $3/8$ (Theorem 1.1). We do not claim the true exponent, and we make no claim that $3/8$ is the best exponent obtainable from this frame or from known techniques. Better constants for Theorem 1.1, and a smaller threshold than $4 \cdot 10^{26}$, would also be of interest. It would also be interesting to have a matching structure for the lower bound, or a direct argument that some end cost of order b^β with $\beta > 0$ is unavoidable in this frame.

APPENDIX A. THE NORMALISED ENCLOSURES OF COROLLARY 8.3

This appendix lists the inequalities evaluated on each box in step (iii) of the proof of Corollary 8.3; they are those of the program `cert3e.py` (functions `box` and `wall`). Put $z := m^{-1/4}$ and $\tau_n := t/z^2$. Enclosures of the inputs: $\hat{b} \in [12^{5/4}(1 - 5.000001z^4)^{5/4}, 12^{5/4}]$, $\mu b \in [2, 2.000001]$, $\Delta \leq 2.000001$, $D_\ell z^3, D_r z^3 \in [4 - z^3, 4]$, $\delta_Q \leq g_q z^5$ with $g_q := 2.1/(16\hat{b})$, and $1/Q_f \leq f_q z^5$ with $f_q := 1/(16\hat{b})$. With $t = \tau_n z^2$:

$$\frac{\tan \alpha}{z^2} = \frac{2\tau_n}{1-t^2}, \quad \frac{\sin \alpha}{z^2} = \frac{2\tau_n}{1+t^2}, \quad q_n := \frac{q}{z^4} = \frac{2\tau_n^2}{1+t^2}, \quad \frac{\sec \alpha - 1}{z^4} = \frac{2\tau_n^2}{1-t^2}, \quad \frac{\kappa_m}{z^2} \leq \frac{q_n^2 \hat{b} z}{\mu b},$$

and $\tau/z^2 \in [\tan \alpha/z^2 - \kappa_m/z^2 - g_q z^3, \tan \alpha/z^2 - \kappa_m/z^2]$.

Main part. (E-i) $1/10 - t > 0$. (E-ii) $z^5(w - D_\ell - D_r - m \tan \alpha - \sec \alpha) \geq \hat{b}(1 - 2.1z^{10}/\hat{b}^2) - (D_\ell z^3 + D_r z^3)z^2 - (\tan \alpha/z^2)z^3 - \sec \alpha z^5 > 0$. (E-iii) $\tan \alpha/z^2 - \kappa_m/z^2 - g_q z^3 > 0$. (E-iv) $1 - \varepsilon_m > 0$ with $\varepsilon_m \leq ((\sec \alpha - 1)/z^4)z^4 + g_q z + f_q z^5$ (together with the direct bound on τ in (iii)). The main terms of Lemma 8.2, times z^3 : $w \sin \alpha z^3 \leq \hat{b} \sin \alpha/z^2$; $wqz^3 \leq \hat{b}q_n z^2$; $\Delta(\sec \alpha + m \tan \alpha)z^3 \leq \Delta(\sec \alpha z^3 + (\tan \alpha/z^2)z)$; $J_0 z^3 \leq (4\tau_n z^2 + 2z^4)/(16\hat{b}q_n)$, times $1 + (\tan \alpha/z^2)z^2$; $J_L(\varepsilon_m + \tan \alpha)z^3 \leq ((\sec \alpha - 1)/z^4)z^3 + g_q + f_q z^4 + (\tan \alpha/z^2)z$; $\frac{1}{2}\mu h_L^2 z^3 \leq \mu b(h_L z^3)^2 z^2/(2\hat{b})$ with $h_L z^3 \in [D_\ell z^3, D_\ell z^3 + 3.000001(\tan \alpha/z^2)z^5]$; and $\mu w_R^2 z^3 \leq \mu b(D_r z^3 + \sec \alpha z^3 + (\tan \alpha/z^2)z)^2 z^2/\hat{b}$.

Walls. A wall with $\zeta_n := \zeta/z^2$, $\chi_n := \chi z^3$ and $L_n := Lz^4 \leq 1$ (upper value 1 used) has $a_n := \tan \alpha_0/z \in [\frac{3}{2}\sqrt{\zeta_n}(1 - \epsilon'), \frac{3}{2}\sqrt{\zeta_n}(1 + \epsilon') + g_q z^4]$, $\sec \alpha_0 = \sqrt{1 + a_n^2 z^2}$, $\cos \alpha_0 = 1/\sec \alpha_0$, $(\sec \alpha_0 - 1)/z^2 = a_n^2/(\sec \alpha_0 + 1)$, $q_{0,n} := q_0/z^2 = a_n^2/(\sec \alpha_0(\sec \alpha_0 + 1))$, $q_{\min,n} := q_{\min}/z^2 = (\varpi a_n)^2/(s_m(s_m + 1))$ with $s_m = \sqrt{1 + \varpi^2 a_n^2 z^2}$, $N_0 z^3 \in [\chi_n - 2z^3, \chi_n - z^3]$, $[\chi]z^3 \in [\chi_n - z^3, \chi_n]$, $r(L)z^3 = \chi_n + \zeta_n L_n z$, $M_{\max} z^3 = r(L)z^3 + N_0 z^3 q_{0,n} z^2$,

$$\Lambda z^3 = \frac{q_{\min,n} N_0 z^3 - z - \zeta_n a_n z^4}{\zeta_n}, \quad \bar{K} z = \frac{L_n}{\Lambda z^3}, \quad \frac{\varepsilon_W}{z^2} = \frac{\sec \alpha_0 - 1}{z^2} + N_0 z^3 g_q + f_q z^3.$$

The hypotheses, normalised: (W1) $\Lambda z^3 - N_0 z^3 a_n z \geq 0$; (W2) $(1 - \varpi)a_n - \Delta_{\text{dr}}/z > 0$ with $\Delta_{\text{dr}}/z = \frac{q_{0,n}}{N_0 z^3}(L_n + \bar{K} z(1 + \zeta_n a_n z^3)/\zeta_n) + \bar{K} z g_q z^3$; (W3) $N_0 z^3 \cos \alpha_0 - z^3(\zeta_n \sec \alpha_0/q_{\min,n} + a_n z + \zeta_n z^2(2 + \sec \alpha_0) + \zeta_n M_{\max} z^3 a_n) - z(1 + \zeta_n a_n z^3)/q_{\min,n} \geq 0$; (W4) $1 - (\varepsilon_W/z^2)z^2 > 0$; (W5) $\varpi a_n - g_q z^4 \geq 0$ and $a_n z \leq 1/10$. The bound: $B_W z^3 \leq \alpha L_n + R$ with $\alpha := c_{\text{col}}/z + X/(\Lambda z^3)$, where $c_{\text{col}}/z = a_n + \sec \alpha_0 q_{0,n} z + \sec \alpha_0^2 \zeta_n z + \sec \alpha_0 \zeta_n N_0 z^3 q_{0,n} a_n z$, $X = (N_0 z^3 - z^3)((\varepsilon_W/z^2)z + a_n) + (1 + \zeta_n a_n z^3)(1 + a_n z)/q_{\min,n} + h_T(w_T z)z$, $h_T = 1 + \zeta_n \sec \alpha_0/q_{\min,n} + a_n z + \zeta_n z^2(1 + \sec \alpha_0) + \zeta_n M_{\max} z^3 a_n$, $w_T z = z + (r(L)z^3 + z^3)(q_{0,n}^2/\zeta_n + g_q z^3)$, and $R = [\chi]z^3(1 + \frac{1}{2}a_n z) + w_0 z^3(1 + \zeta_n z^2 w_0) + r(L)z^3(1 + \frac{1}{2}a_n z) + (w_e z^2)z(4 + 2\zeta_n w_e z^2)$ with $w_0 = a_n z + N_0 z^3 q_{0,n} a_n$ and $w_e z^2 = z^2(1 + \sec \alpha_0) + r(L)z^3 a_n + N_0 z^3 q_{0,n} a_n z^2$.

The three walls use: Z_L : $\zeta_n = \tan \alpha/z^2$, $\chi_n \in [D_\ell z^3, D_\ell z^3 + 3.000001(\tan \alpha/z^2)z^5]$; Z_{LR} : $\zeta_n = \tau/z^2$, $\chi_n \in [D_r z^3 + ((\sec \alpha - 1)/z^4)z^7, D_r z^3 + 2 \sec \alpha z^3]$; Z_{UR} : $\zeta_n = \tan \alpha/z^2$, $\chi_n \in [D_r z^3, D_r z^3 + \sec \alpha z^3 + (\tan \alpha/z^2)z]$. Finally $\Psi \leq w \sin \alpha z^3 + (\text{other main terms}) + (\alpha + R)_{Z_L} + \max(\alpha_{Z_{LR}}, \alpha_{Z_{UR}}) + R_{Z_{LR}} + R_{Z_{UR}}$.

VERIFICATION STATUS

- Theorems 1.3 and 1.4, Corollary 5.1, Proposition 1.5 and Lemmas 2.1–4.3, 6.1, Corollary 6.2 and Proposition 6.3 are proved in full in this text. An earlier write-up of Sections 3–5 received an independent AI review, which found no critical or major issue; its minor remarks are included (the case $(k, b) = (2, 2)$, $L \geq 0$ in Lemma 3.4, and the scope of Lemma 4.3, see Remark 5.3). The formulas were also tested in exact arithmetic: (5.1) and the bounds of Theorems 1.3, 1.4 and Proposition 1.5 for all $2 \leq b \leq k \leq 1500$, the angle of Lemma 4.2 for all $2 \leq b \leq k \leq 250$, and actual packings (separating axis tests) for all b and $k \leq 60$ and some larger cases. Such tests are not proofs.
- Theorem 1.2 (Section 7) is proved in full; it uses Lemma 2.2, Lemmas 3.1–3.3, Lemma 6.1, Corollary 6.2 (with Lemma 4.1) and Proposition 6.3. An earlier write-up of Section 7 received two independent blank-slate AI reviews, both of which judged it correct with no critical or major issue; their minor remarks are included. The proof relies on one evaluation in interval arithmetic, $\Phi(10^{-4/3}) \leq 5.0163$, done by the author’s program and by three independent computations; the other constants were certified in the same way. The construction was tested in exact arithmetic, also by an independent reimplementaion (Remark 7.5). To our knowledge the proof has not appeared before; it deserves priority in a human review.
- Theorem 1.1 (Section 8) is proved in full; it uses Lemmas 7.1 and 7.2, Lemma 2.2, Lemmas 3.1–3.3, Lemma 6.1, Corollary 6.2 and Proposition 6.3. An earlier write-up of Section 8 received two independent blank-slate AI reviews, with their own programs: one judged it correct after fixes, the other correct; all their fixes are included (the hypothesis (E-iv) in the form $\varepsilon_m + \tau < 1$, two table values, the decomposition of the certified bound, and the reason for $b_0 = 10^{16}$). An independent reimplementaion found no counterexample. The author’s own interval program for Corollary 8.3 used too little slack on the wall tilts; the constant $C_E = 15.54$ is certified by three independent full-range interval coverings ($C_E \leq 14.964$, 14.963 and 14.9631; `mpmath.iv` twice and `python-flint arb` once). The asymptotic regime ($b \geq 10^{16}$) cannot be tested by explicit computation; the explicit checks cover the validity of the construction at smaller b (Remark 8.4). It deserves priority in a human review.
- Theorem 1.6 is computer-assisted. Its finite part was checked by three programs, two of them written independently of the generating program (Section 9).
- Section 12 contains remarks only; Remarks 12.1 and 12.2 are not proved here.
- Section 10 and the novelty statement rest on a literature search, which cannot prove absence.
- No human expert has reviewed this text. The paper has not been refereed. All reviewers and checkers so far are AI systems of the same family, which may share blind spots.

Programs and data. <https://github.com/squarepacker/k2-minus-c-upper> (archived on Zenodo; the DOI is given in the repository README). The data files `stair_k*.json.gz` (the end pieces of the five certificates), the checker `stair_check.py`, the generator `stair_dump.py` and the interval computation `const_stair.py` of the constants of Corollary 7.4 and Section 7.3, the interval covering `cert3e.py` of the constants of Corollary 8.3, and the wall filler `r38.py` with the exact checker `chk.py` and the driver `run_z.py` (Remark 8.4) accompany this text; `README_code.txt` lists the commands and the expected outputs.

Use of AI. Developed with the assistance of Claude (Anthropic), including the proofs, the text and the programs; the author takes full responsibility.

Licence. This text is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0); the accompanying programs are distributed under the MIT licence.

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